

[MS-UNMP]:

User Name Mapping Protocol

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1 Introduction

The Windows and **UNIX** operating systems use different mechanisms for user identification, authentication, and resource access control. Users have separate accounts in the Windows portion and the UNIX portion of any network. Because Windows and UNIX user identifications and user names are stored and used differently, there is no association between the two sets, even though the same users exist on each network.

The User Name Mapping Protocol maps Windows domain user and group account names (DOMAIN\NAME) to the **POSIX** user and group identifiers (UIDs and **GIDs**) utilized in AUTH_UNIX authentication and vice versa. This enables the association of user names for users who have different identities in Windows-based and UNIX-based domains. For example, this protocol allows user and group accounts from multiple Windows domains to access resources on Network File System (NFS) file servers by using UIDs and GIDs. The User Name Mapping Protocol supports only retrieval of mappings; it does not include procedures for changing user mappings.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

advanced map: Used to map accounts that have different names on the **UNIX** and Windows systems. **Advanced maps** are also used to map users from different Windows domains, and they can also explicitly map accounts that would generally be mapped by **simple maps**. For more information, see [\[NFSAUTH\]](#).

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a domain controller (DC) and host a member list that identifies all members of the domain, as well as optionally hosting the Active Directory service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

DUMPALLMAPSXXX_PROC: A reference to the following procedures: DUMPALLMAPS_PROC, DUMPALLMAPSEX_PROC, DUMPALLMAPSW_PROC, and DUMPALLMAPSEXW_PROC.

group identifier (group ID or GID): A number that identifies a group of users to a **UNIX** operating system. The scope of the number is at least machine-wide but can also be coordinated across a group of machines by means of services, such as the Network Information Service (NIS).

group map: An association between a Windows group account name, a **UNIX** group account name, and a **GID**.

little-endian: Multiple-byte values that are byte-ordered with the least significant byte stored in the memory location with the lowest address.

map: An association between a Windows-based network user or group name and a **UNIX**-based network user or group name.

multibyte character set (MBCS): An alternative to **Unicode** for supporting character sets, like Japanese and Chinese, that cannot be represented in a single byte. Under MBCS, characters are encoded in either one or two bytes. In two-byte characters, the first byte, or "lead" byte, signals that both it and the following byte are to be interpreted as one character. The first byte comes from a range of codes reserved for use as lead bytes. Which ranges of bytes can be lead bytes

depends on the code page in use. For example, Japanese code page 932 uses the range 0x81 through 0x9F as lead bytes, but Korean code page 949 uses a different range.

Network File System (NFS): A **Network File System** protocol, as specified in [\[RFC1094\]](#) and [\[RFC1813\]](#). This protocol is compatible with **NFS** version 3 (NFSv3). **NFS** version 4 (NFSv4) obviates the need for this protocol by allowing Windows and **UNIX domains** to interoperate using Kerberos version 5, which allows them to share the same namespace.

OEMCP: The default OEM code page of the system. The OEM code page is used for conversions of MS-DOS-based, text-mode applications.

portmapper service: A **portmapper service** is a **SUNRPC** service that provides discovery services; clients of the **portmapper service** can use it to discover other **SUNRPC** services running on the same computer. The information returned by the **portmapper service** is then used by the client of the **portmapper service** to act as a client for the discovered **SUNRPC** service. The **portmapper service** runs on a specific well-known port (Port 111 on TCP/UDP).

POSIX: Portable Operating System Interface, as specified in [\[IEEE1003.1\]](#). **POSIX** is a set of standard operating system interfaces based on **UNIX**. This term is used interchangeably with **UNIX** in the rest of this document, as described in [\[IEEE1003.1\]](#).

primary map: When multiple Windows accounts are mapped to a single **UNIX** account, one of these **mappings** can be designated as a "primary" **mapping**. For more information, see [\[NFSAUTH\]](#).

security identifier (SID): An identifier for security principals that is used to identify an account or a group. Conceptually, the **SID** is composed of an account authority portion (typically a **domain**) and a smaller integer representing an identity relative to the account authority, termed the relative identifier (RID). The **SID** format is specified in [\[MS-DTYP\]](#) section 2.4.2; a string representation of **SIDs** is specified in [\[MS-DTYP\]](#) section 2.4.2 and [\[MS-AZOD\]](#) section 1.1.1.2.

simple map: **Maps** between accounts with the exact same name in **UNIX** as in Windows. For more information, see [\[NFSAUTH\]](#).

SUNRPC: A remote procedure call (RPC) protocol from Sun Microsystems [\[RFC1057\]](#). **SUNRPC** forms the basis of the **Network File System (NFS)** Protocol. **SUNRPC** has no relationship to Remote Procedure Call Protocol Extensions, as specified in [\[MS-RPCE\]](#).

Unicode: A character encoding standard developed by the Unicode Consortium that represents almost all of the written languages of the world. The **Unicode** standard [\[UNICODE5.0.0/2007\]](#) provides three forms (UTF-8, UTF-16, and UTF-32) and seven schemes (UTF-8, UTF-16, UTF-16 BE, UTF-16 LE, UTF-32, UTF-32 LE, and UTF-32 BE).

UNIX: A multiuser, multitasking operating system developed at Bell Laboratories in the 1970s. In this document, the term "**UNIX**" is used to refer to any derivatives of this operating system.

user identifier (user ID or UID): A number that identifies a particular user to a **UNIX** operating system. The scope of the number is at least machine-wide and can be coordinated across a group of machines by means of services such as NIS.

user map: An association between a Windows user account name, a **UNIX** user account name, and a UID.

wide characters: Characters represented by a 2-byte value that are encoded using **Unicode** UTF-16. Unless otherwise stated, no range restrictions apply.

XDR: The data encoding standard used by **SUNRPC** for a selection of common data types such as strings, integers, and arrays of integers, as specified in [\[RFC4506\]](#).

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[IEEE1003.1] The Open Group, "IEEE Std 1003.1, 2004 Edition", 2004, http://www.unix.org/version3/ieee_std.html

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[RFC1057] Sun Microsystems, Inc., "RPC: Remote Procedure Call Protocol Specification Version 2", RFC 1057, June 1988, <https://www.rfc-editor.org/info/rfc1057>

[RFC1831] Srinivasan, R., "RPC: Remote Procedure Call Protocol Specification Version 2", RFC 1831, August 1995, <https://www.rfc-editor.org/info/rfc1831>

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC4506] Network Appliance, Inc., "XDR: External Data Representation Standard", STD 67, RFC 4506, May 2006, <https://www.rfc-editor.org/info/rfc4506>

1.2.2 Informative References

[MSFT-ADNaming] Microsoft Corporation, "Naming conventions in Active Directory for computers, domains, sites, and OUs", <https://learn.microsoft.com/en-US/troubleshoot/windows-server/identity/naming-conventions-for-computer-domain-site-ou>

[MSFT-SFUPerms] Microsoft Corporation, "Permissions In Microsoft Services for UNIX v3.0", <https://technet.microsoft.com/en-us/library/bb463216.aspx>

[NFSAUTH] Russel, C., "NFS Authentication", <https://technet.microsoft.com/en-us/library/bb463203.aspx>

[NIS] Sun Microsystems, Inc., "System Administration Guide: Naming and Directory Services (DNS, NIS, and LDAP)", <https://docs.oracle.com/cd/E19253-01/816-4556/>

1.3 Overview

The User Name Mapping Protocol maps Windows domain identities (user and group account names) to **UNIX** user and UNIX group identities (user and group account names and their corresponding UID and **GID**) and vice versa. Clients of the User Name Mapping Protocol use **SUNRPC**-formatted messages to enumerate and/or translate user and group account information between a UNIX and a Windows domain. The User Name Mapping Protocol exists to allow a one-to-one mapping of each Windows group account name to a GID number and a one-to-one mapping of each Windows user account name to a **user identifier (user ID or UID)** number.

The User Name Mapping Protocol is invoked by a client application when the application needs to provide a **user map** or a **group map** between a UNIX user or group and the corresponding Windows user or group. This need is application specific and is not specified by the User Name Mapping Protocol. The UNIX or Windows user, or UNIX or Windows group, that needs to be mapped is supplied to the User Name Mapping Protocol by the client application, and the mapped user/group is returned to the client by the User Name Mapping Protocol server. For user mapping and group mapping enumerations, the client application specifies the enumeration parameters, and the User Name Mapping Protocol server returns the enumerated user mappings/group mappings to the client.

These mapping enumerations can be cached by the client application and kept up to date by periodically polling the server to determine if the cached mappings are still valid. The User Name Mapping Protocol does not provide authentication or authorization of the application-provided user/group; to the client, it is a read-only account mapping service.

An example of this authentication behavior is a user on a UNIX machine making a file access request that contains AUTH_UNIX-formatted user credentials to an **NFS** server implemented on a computer running Windows. The NFS server acts as a User Name Mapping Protocol client (or "user map") to request the Windows domain user and group names (from the User Name Mapping Protocol server) that match the AUTH_UNIX credentials, [\[RFC1057\]](#) section 9.2, supplied by the UNIX user. This action enables the NFS server to authenticate the file access request.

This document specifies the SUNRPC-formatted messages that provide support for the following operations:

- Mapping **POSIX** user and group names and/or UIDs/GIDs to Windows domain and account names.
- Mapping Windows domain and account names to POSIX user and group names, and UIDs and GIDs.
- Allowing a User Name Mapping Protocol client to authenticate a POSIX user by providing a user name and password.
- Enumerating all user mappings and group mappings between POSIX accounts and Windows accounts known to the User Name Mapping Protocol server.
- Testing to see if any maps previously enumerated by a client have changed from the time of the last check.
- Mapping a Windows domain **security identifier (SID)** to a POSIX user/group name and UID/GID.

This document specifies versions 1 and 2 of the User Name Mapping Protocol. Version 1 is comprised of a set of nine SUNRPC procedures; version 2 consists of a set of 18 SUNRPC procedures. For a list of these procedures, see the table in section [2.2.5](#).

There are several differences between User Name Mapping Protocol version 1 and User Name Mapping Protocol version 2. Version 2 added procedures 10–17, which are the **wide character (Unicode)** counterparts of procedures 1–4 and 6–9. Procedures 1–4 and 6–8 accept **multibyte character set (MBCS)** character-encoded strings as input. Version 2 includes the additional procedure 9, which takes a Windows account SID and returns an MBCS character-encoded UNIX account **map** that corresponds to the Windows account represented by that SID. The wide character (Unicode) counterpart to procedure 9 is procedure 17.

1.4 Relationship to Other Protocols

The User Name Mapping Protocol relies on [\[RFC1057\]](#) and [\[RFC4506\]](#) for communicating with clients by means of the **SUNRPC** message version 2 and **XDR** protocols as specified in those documents. The User Name Mapping Protocol uses SUNRPC authentication level AUTH_NULL (as specified in

[RFC1057]). The User Name Mapping Protocol uses SUNRPC message version 2 implemented on top of TCP and UDP. The User Name Mapping Protocol message formats are not sensitive to which underlying transports (TCP or UDP) are being used.

1.5 Prerequisites/Preconditions

It is required that the User Name Mapping Protocol server has been previously configured with all the appropriate **UNIX** and Windows domain name and group mapping information, and that it has registered with the **portmapper service** (as specified in [\[RFC1057\]](#) Appendix A) on the same computer as the User Name Mapping Protocol server. Normal TCP/IP services sufficient to provide TCP-based or UDP-based communications must be available.[<1>](#)

1.6 Applicability Statement

The User Name Mapping Protocol is applicable in a heterogeneous network environment where users have separate accounts in **UNIX** and Windows infrastructures. This protocol provides a means to associate user and group accounts in two networks for users or groups that have different identities in UNIX-based and Windows-based administrative **domains**.

1.7 Versioning and Capability Negotiation

This document covers versioning issues in the following areas:

- **Protocol Versions:** The User Name Mapping Protocol supports versions 1 and 2. These dialects are defined in section [2.2](#).
- **Capability Negotiation:** Version negotiation of the User Name Mapping Protocol is achieved using the standard method for protocol negotiation for **SUNRPC** services as specified in [\[RFC1057\]](#) section 8. The User Name Mapping Protocol client requests a specific version of the User Name Mapping Protocol from the **portmapper service** (as specified in [\[RFC1057\]](#) Appendix A). The portmapper service replies with the available versions registered by the User Name Mapping Protocol server. It is recommended that requests made to the User Name Mapping Protocol server for versions other than those supported with a SUNRPC PROG_MISMATCH message, as specified in [\[RFC1057\]](#).

1.8 Vendor-Extensible Fields

None.

1.9 Standards Assignments

Parameter	Value	Reference
MAPSVC_PROGRAM	351455	[RFC1057] section 7.3

2 Messages

2.1 Transport

The User Name Mapping Protocol is a **SUNRPC** protocol (as specified in [\[RFC1057\]](#)) that runs on TCP/IP using TCP and/or UDP transports, with a well-known program number of MAPSVC_PROGRAM (351455). The User Name Mapping Protocol server registers an available TCP/UDP port with the local **portmapper service** on startup using the MAPSVC_PROGRAM number for all combinations of TCP, UDP, and protocol versions that the User Name Mapping Protocol server is capable of or configured to accept. User Name Mapping Protocol clients query the SUNRPC portmapper server for the TCP/UDP port number on which the User Name Mapping Protocol is registered and listening for the requested version and transport combination.

Configuration of the portmapper service and port registration is specified in [\[RFC1057\]](#) Appendix A. The User Name Mapping Protocol does not define a configuration interface to the portmapper service.

The User Name Mapping Protocol server provides a procedure-oriented interface to the User Name Mapping Protocol clients. Clients identify the remote procedure by using a combination of a 32-bit program number, a 32-bit version number, and a 32-bit procedure number (as specified in [\[RFC1057\]](#)). The service is stateless; every SUNRPC call is self-contained and does not depend on the previous calls made or previous state of the service.

The User Name Mapping Protocol server accepts all SUNRPC packets with an authentication level of AUTH_NULL, as specified in [\[RFC1057\]](#) section 9.1. Therefore, no authentication information is required by the client.

2.2 Message Syntax

The following structures are specified in **XDR** Data Definition Language syntax (as specified in [\[RFC4506\]](#) section 6) while procedures are defined in the **SUNRPC** language, as specified in [\[RFC1057\]](#) section 11.

2.2.1 User Name Mapping Protocol Message Headers

2.2.1.1 SUNRPC Request Header

The User Name Mapping Protocol uses standard **SUNRPC** version 2 msg_type CALL headers. Requests are made with an authentication level of AUTH_NULL. This header format and its fields and values are specified in [\[RFC1057\]](#) section 8.

2.2.1.2 SUNRPC Response Header

The User Name Mapping Protocol uses standard **SUNRPC** version 2 msg_type REPLY headers. This header format and its fields and values are specified in [\[RFC1057\]](#) section 8.

2.2.2 Common User Name Mapping Protocol Data Types

In this section, the **XDR** Data Description Language (as specified in [\[RFC4506\]](#)) is used to specify the XDR format parameters and results of each of the **SUNRPC** service procedures that a User Name Mapping Protocol server provides.

2.2.2.1 Sizes

```
const MAXNAMELEN = 128;  
const MAXNAMELENx2 = 256;
```

```
const MAXLINELEN = 256;
const MAXLINELENx2 = 512;
const MAXGIDS = 32;
const MAXSIDLEN = 72;
```

(MAXGIDS is the maximum count of **GIDs**. This includes both the primary GID and any supplementary GIDs.)

2.2.2.2 MapSvrMBCSNameString

```
typedef opaque MapSvrMBCSNameString<MAXNAMELEN>;
```

An **XDR** variable-length opaque data field, as specified in [\[RFC4506\]](#) section 4.10, whose maximum length is specified in bytes. The length is equal to the number of **MBCS** bytes encoded in the system **OEM code page (OEMCP)**, including multibyte characters, as specified by the **length** field that precedes the byte stream. The value of the **length** field MUST NOT exceed the value MAXNAMELEN. Minimum length is 0.

2.2.2.3 MapSvrUnicodeNameString

```
typedef opaque MapSvrUnicodeNameString<MAXNAMELENx2>;
```

An **XDR** variable-length opaque data field, as specified in [\[RFC4506\]](#) section 4.10, whose maximum length is specified in bytes. The maximum length is defined by the **length** field that precedes the byte stream. The value of the **length** field MUST NOT exceed the value MAXNAMELENx2. The maximum length of the character string is equal to as many 2-byte **Unicode** (UTF-16) characters as can be stored in a MapSvrUnicodeNameString, with a maximum length equal to **length**. Minimum length is 0.

2.2.2.4 MapSvrMBCSWindowsNameString

```
typedef opaque MapSvrMBCSWindowsNameString<MAXLINELEN>;
```

An **XDR** variable-length opaque data field, as specified in [\[RFC4506\]](#) section 4.10, whose maximum length is specified in bytes. The length is equal to the number of **MBCS** bytes encoded in the system **OEMCP**, including multibyte characters, as specified by the **length** field that precedes the byte stream. The value of the **length** field MUST NOT exceed the value MAXLINELEN. Minimum length is 0.

2.2.2.5 MapSvrUnicodeWindowsNameString

```
typedef opaque MapSvrUnicodeWindowsNameString<MAXLINELENx2>;
```

An **XDR** variable-length opaque data field, as specified in [\[RFC4506\]](#) section 4.10, whose maximum length is specified in bytes. The maximum length is defined by the **length** field that precedes the byte stream. The value of the **length** field MUST NOT exceed the value MAXLINELENx2. The maximum length of the character string is equal to as many 2-byte **Unicode** (UTF-16) characters as can be stored in a MapSvrUnicodeWindowsNameString, with a maximum length equal to **length**. Minimum length is 0.

2.2.2.6 MapSvrMBCSMapString

```
typedef opaque MapSvrMBCSMapString<MAXLINELEN>;
```

An **XDR** variable-length opaque data field, as specified in [\[RFC4506\]](#) section 4.10, whose maximum length is specified in bytes. The length is equal to the number of **MBCS** bytes encoded in the system **OEMCP**, including multibyte characters, as specified by the **length** field that precedes the byte stream. The value of the **length** field MUST NOT exceed the value MAXLINELEN. Minimum length is 0.

This type is used to define a single account **map** as a colon-delimited string of MBCS characters. This type is returned as an output from the map enumeration procedure. For more details, see section [2.2.3.2](#).

The format of MapSvrMBCSMapString is a sequence of colon-delimited fields. It has one of two forms, depending on the context: **user map** or **group map**, as follows.

For user map, MapSvrMBCSMapString has the following format.

```
MapType:WindowsAccountName:AuthType:UNIXDomain:UNIXServer:
UNIXAccountName:UNIXPassword:ID:GIDArray
```

For group map, MapSvrMBCSMapString has the following format.

```
MapType:WindowsAccountName:AuthType:UNIXDomain:UNIXServer:
UNIXAccountName:GID
```

MapType: A single MBCS character that indicates the type of map from which the mapping was derived. It MUST be one of the following characters.

Value	Meaning
'*'	The map is a primary map .
'^'	The map is an advanced map .
'_'	The map is a simple map .

WindowsAccountName: A string of MBCS characters that contains the Windows account name. It MUST be in DOMAIN\NAME format.

AuthType: A single MBCS character that indicates which entity provided the map. **AuthType** MUST be one of the values in the following table. If the value is AUTH_NIS, the source MUST be a NIS service on the network. If the value is AUTH_FILE, the source SHOULD [<2>](#) be from the service-maintained database local to the User Name Mapping Protocol server.

Value	Meaning
'0' (AUTH_FILE)	The map was obtained from a service-maintained database local to the User Name Mapping Protocol server. The form of the database is implementation-specific.
'1' (AUTH_NIS)	The map was obtained from a NIS service on the network. NIS is specified in [NIS] .

UNIXDomain: A string of MBCS characters that contains the string "PCNFS" if the map was obtained from a service-maintained database, or the NIS server domain to which the account belongs if the map was obtained from a NIS service. If **AuthType** is equal to AUTH_NIS, this field MUST contain the NIS server domain the account is a member of.

UNIXServer: A string of MBCS characters that contains the string "PCNFS" if the map was obtained from a service-maintained database, or a string that represents the NIS server name to which the account belongs if the map was obtained from a NIS server.

UNIXAccountName: A string of MBCS characters that represents the **UNIX** account name.

UNIXPassword: A sequence of bytes that represents the password for a user record as returned by a call to the crypt() API that uses the user's cleartext password, as specified in [\[IEEE1003.1\]](#) System Interfaces Volume (XSH). This field is empty when the password is not available or does not apply. The password record MUST NOT contain any MBCS colon characters.

ID: A string of MBCS characters that contains the ID for the UNIX account.

GID: A string of MBCS characters that contains the GID for the UNIX account.

GIDArray: A string of MBCS characters that contains the primary and supplementary **GIDs** for the UNIX account, with each supplementary GID after the primary GID, and separated by additional colon characters.

2.2.2.7 MapSvrUnicodeMapString

```
typedef opaque MapSvrUnicodeMapString<MAXLINELENx2>;
```

An **XDR** variable-length opaque data field, as specified in [\[RFC4506\]](#) section 4.10, whose maximum length is specified in bytes. The maximum length is defined by the **length** field that precedes the byte stream. The value of the **length** field MUST NOT exceed the value MAXLINELENx2. The maximum length of the character string is equal to as many 2-byte **Unicode** (UTF-16) characters as can be stored in a MapSvrUnicodeMapString, with a maximum length equal to **length**. Minimum length is 0.

This type is used to define a single account map in colon-delimited string format when returned as an output from the map enumeration procedure. For more details, see section [2.2.3.4](#).

The format of a MapSvrUnicodeMapString field is a sequence of colon-delimited fields as specified in section [2.2.2.6](#), substituting Unicode characters for **MBCS** characters.

2.2.2.8 unix_account

This type is used to specify a **UNIX** account name in **MBCS** format, in addition to an ID used to search for the corresponding Windows account information when mapping a UNIX account name to a Windows account name. For more information, see sections [2.2.5.2](#) and [2.2.5.8](#).

```
struct unix_account {  
    long SearchOption;  
    long Reserved;  
    long ID;  
    MapSvrMBCSNameString UnixAccountName;  
};
```

SearchOption: An **XDR**-encoded, 32-bit signed integer that defines the user search criteria to use for the request. **SearchOption** MUST be one of the following values.

Value	Meaning
0x00000001	If set, UnixAccountName is valid and MUST be used as the search criterion.
0x00000002	If set, ID is valid and MUST be used as the search criterion.
0x00000003	If set, UnixAccountName and ID are both valid and both MUST be used as the search criteria.

Reserved: A 32-bit signed integer that MUST be sent as 0x00000000 and MUST be ignored on receipt.

ID: An XDR-encoded, 32-bit signed integer that contains the UNIX account ID to use as the search criterion. If **SearchOption** is not 0x00000002 or 0x00000003, this value MUST be ignored.

UnixAccountName: A [MapSvrMBCSNameString \(section 2.2.2.2\)](#) that contains the name of the UNIX account to use as the search criterion. The length of the string MUST NOT exceed 128 bytes. POSIX user and group account name constraints are specified in [\[IEEE1003.1\]](#). If **SearchOption** is not 0x00000001 or 0x00000003, this value MUST be ignored.

2.2.2.9 unix_accountW

This type is used to specify a **UNIX** account name in **Unicode** format, in addition to an ID used to search for the corresponding Windows account information when mapping a UNIX account name to a Windows account name. For more information, see sections [2.2.5.13](#) and [2.2.5.16](#).

```
struct unix_accountW {  
    long SearchOption;  
    long Reserved;  
    long ID;  
    MapSvrUnicodeNameString UnixAccountName;  
};
```

SearchOption: An **XDR**-encoded, 32-bit signed integer that defines the user search criteria to use for the request. **SearchOption** MUST be one of the following values.

Value	Meaning
0x00000001	If set, UnixAccountName is valid and MUST be used as the search criterion.
0x00000002	If set, ID is valid and MUST be used as the search criterion.
0x00000003	If set, UnixAccountName and ID are both valid and both MUST be used as the search criteria.

Reserved: A 32-bit signed integer that MUST be sent as 0x00000000 and MUST be ignored on receipt.

ID: An XDR-encoded, 32-bit signed integer that contains the UNIX account ID to use as the search criterion. If **SearchOption** is not 0x00000002 or 0x00000003, this value MUST be ignored.

UnixAccountName: A [MapSvrUnicodeNameString \(section 2.2.2.3\)](#) that contains the name of the UNIX account to use as the search criterion. The length of the string MUST NOT exceed 256 bytes. POSIX user and group account name constraints are specified in [\[IEEE1003.1\]](#). If **SearchOption** is not 0x00000001 or 0x00000003, this value MUST be ignored.

2.2.2.10 unix_user_auth

This type is used to specify a **UNIX** account name (in **MBCS** format) and a password to retrieve the set of UNIX account details that correspond to the account. For more details, see section [2.2.5.4](#).

```
struct unix_user_auth {  
    MapSvrMBCSNameString UnixUserAccountName;  
    MapSvrMBCSNameString UnixUserAccountPassword;  
};
```

UnixUserAccountName: A [MapSvrMBCSNameString \(section 2.2.2.2\)](#) that contains the name of the UNIX user account to use as the search criterion. The length of this string MUST NOT exceed 128 bytes. POSIX user and group account name constraints are specified in [\[IEEE1003.1\]](#).

UnixUserAccountPassword: An **XDR** variable-length opaque data field, as defined in [\[RFC4506\]](#) section 4.10, that contains the password of the UNIX user account to use as the search criterion. The length of this field MUST NOT exceed 128 bytes. This string MUST be generated by a call to the **POSIX** crypt function, as specified in [IEEE1003.1] System Interfaces Volume (XSH) Chapter 3.

2.2.2.11 unix_user_authW

This type is used to specify a **UNIX** account name (in **Unicode** format) and a password to retrieve the set of UNIX account details that correspond to the account. For more details, see section [2.2.5.15](#).

```
struct unix_user_authW {
    MapSvrUnicodeNameString UnixUserAccountName;
    MapSvrUnicodeNameString UnixUserAccountPassword;
};
```

UnixUserAccountName: A [MapSvrUnicodeNameString \(section 2.2.2.3\)](#) that contains the name of the UNIX user to use as the search criterion. The length of the string MUST NOT exceed 256 bytes. POSIX user and group account name constraints are specified in [\[IEEE1003.1\]](#).

UnixUserAccountPassword: A [MapSvrUnicodeNameString \(section 2.2.2.3\)](#) that contains the password of the UNIX user account to use as the search criterion. The length of the string MUST NOT exceed 256 bytes. This string MUST be generated by a call to the **POSIX** crypt function, as specified in [IEEE1003.1] System Interfaces Volume (XSH) Chapter 3.

2.2.2.12 windows_creds

This type represents the Windows account name (in **MBCS** format) when used as an output parameter from a search for the corresponding **UNIX** account name (in MBCS format) and/or UNIX ID. For more information, see sections [2.2.5.2](#) and [2.2.5.8](#))

```
struct windows_creds {
    long Status;
    long Reserved;
    MapSvrMBCSWindowsNameString WindowsAccountName;
};
```

Status: An **XDR**-encoded, Boolean return value. This MUST be either 0 or 1. A value of 0 indicates success; a value of 1 indicates failure.

Reserved: A 32-bit signed integer that MUST be 0x00000000 and MUST be ignored on receipt.

WindowsAccountName: A [MapSvrMBCSWindowsNameString \(section 2.2.2.4\)](#) that contains the name of the mapped Windows user or group account that MUST be in the form "DOMAIN\NAME". The length of the string MUST NOT exceed 256 bytes. Windows user/group account name constraints are specified in [\[MSFT-SFUPerms\]](#), and Windows domain naming conventions are specified in [\[MSFT-ADNaming\]](#). If **Status** does not equal 0x00000000, this value MUST be ignored.

2.2.2.13 windows_credsW

This type represents the Windows account name (in **Unicode** format) when used as an output parameter from a search for the corresponding **UNIX** account name (in Unicode format) and/or UNIX ID. For more details, see sections [2.2.5.13](#) and [2.2.5.16](#).

```
struct windows_credsW {
    long Status;
    long Reserved;
    MapSvrUnicodeWindowsNameString WindowsAccountName;
};
```

Status: An **XDR**-encoded, Boolean return value. This MUST be either 0 or 1. A value of 0 indicates success; a value of 1 indicates failure.

Reserved: A 32-bit signed integer that MUST be 0x00000000 and MUST be ignored on receipt.

WindowsAccountName: A [MapSvrUnicodeWindowsNameString \(section 2.2.2.5\)](#) that contains the name of the mapped Windows user or group account that MUST be in the form "DOMAIN\NAME". The length of the string MUST NOT exceed 512 bytes. Windows user account and group account name constraints are specified in [\[MSFT-SFUPerms\]](#), and Windows domain naming conventions are specified in [\[MSFT-ADNaming\]](#). If **Status** does not equal 0x00000000, this value MUST be ignored.

2.2.2.14 windows_account

This type is used to specify a Windows account name in **MBCS** format used to search for the corresponding **UNIX** account information. For more details, see sections [2.2.5.3](#) and [2.2.5.9](#).

```
struct windows_account {
    MapSvrMBCSNameString WindowsAccountName;
};
```

WindowsAccountName: A [MapSvrMBCSNameString \(section 2.2.2.2\)](#) that MUST contain the name of the Windows account in the "DOMAIN\NAME" format to use as the search criterion. Windows user account and group account name constraints are specified in [\[MSFT-SFUPerms\]](#), and Windows domain naming conventions are specified in [\[MSFT-ADNaming\]](#). The length of the string MUST NOT exceed 256 bytes.

2.2.2.15 windows_accountW

This type is used to specify a Windows account name in **Unicode** format used to search for the corresponding **UNIX** account information. For more details, see sections [2.2.5.14](#) and [2.2.5.17](#).

```
struct windows_accountW {
    MapSvrUnicodeNameString WindowsAccountName;
};
```

WindowsAccountName: A [MapSvrUnicodeNameString \(section 2.2.2.3\)](#) that contains the name of the Windows user account to use as the search criterion. The account name MUST be in the form "DOMAIN\NAME". Windows user account and group account name constraints are specified in [\[MSFT-SFUPerms\]](#), and Windows domain naming conventions are specified in [\[MSFT-ADNaming\]](#). The length of the string MUST NOT exceed 512 bytes.

2.2.2.16 unix_auth

This type is used to specify **UNIX** account details returned as a result of an authentication operation on the server. For more details, see sections [2.2.5.4](#) and [2.2.5.15](#).

```
struct unix_auth {
    MapSvrMBCSNameString UnixAccountPassword;
    long ID;
    long GIDArray<MAXGIDS>;
};
```

UnixAccountPassword: A [MapSvrMBCSNameString \(section 2.2.2.2\)](#) that contains the password of the mapped UNIX account. The length of the string MUST NOT exceed 128 bytes.

ID: An **XDR**-encoded, 32-bit signed integer that contains the UNIX **user ID** for the **UnixAccountPassword** that was looked up.

GIDArray: An array of XDR-encoded, 32-bit signed integers that contains the group IDs for the **UnixAccountPassword** that was looked up. The maximum size of this array is MAXGIDS.

2.2.2.17 unix_authW

This type is used to specify **UNIX** account details returned as a result of an authentication operation on the server. For more details, see sections [2.2.5.4](#) and [2.2.5.15](#).

```
struct unix_authW {
    MapSvrUnicodeNameString UnixAccountPassword;
    long ID;
    long GIDArray<MAXGIDS>;
};
```

UnixAccountPassword: A [MapSvrUnicodeNameString \(section 2.2.2.3\)](#) that contains the password of the mapped UNIX account. The length of the string MUST NOT exceed 256 bytes.

ID: An **XDR**-encoded, 32-bit signed integer that contains the UNIX **user ID** for the **UnixAccountPassword** that was looked up.

GIDArray: An array of XDR-encoded, 32-bit signed integers that contains the group IDs for the **UnixAccountPassword** that was looked up. The maximum size of this array is MAXGIDS.

2.2.2.18 unix_creds

This type is used to specify **UNIX** account details returned as a result of a lookup operation on the server. For more details, see sections [2.2.5.3](#), [2.2.5.9](#), and [2.2.5.10](#).

```
struct unix_creds {
    MapSvrMBCSNameString UnixAccountName;
    long ID;
    long GIDArray<MAXGIDS>;
};
```

UnixAccountName: A [MapSvrMBCSNameString \(section 2.2.2.2\)](#) that contains the name of the mapped UNIX account. The length of the string MUST NOT exceed 128 bytes.

ID: An **XDR**-encoded, 32-bit signed integer that contains the UNIX **user ID** for **UnixAccountName**.

GIDArray: An array of XDR-encoded, 32-bit signed integers that contains the group IDs for **UnixAccountName**. The maximum size of this array is MAXGIDS.

2.2.2.19 unix_credsW

This type is used to specify **UNIX** account details returned as a result of a lookup operation on the server. For more details, see sections [2.2.5.14](#), [2.2.5.17](#), and [2.2.5.18](#).

```
struct unix credsW {
    MapSvrUnicodeNameString UnixAccountName;
    long ID;
    long GIDArray<MAXGIDS>;
};
```

UnixAccountName: A [MapSvrUnicodeNameString](#) (section 2.2.2.3) that contains the name of the mapped UNIX account. The length of the string MUST NOT exceed 256 bytes.

ID: An **XDR**-encoded, 32-bit signed integer that contains the UNIX **user ID** for **UnixAccountName**.

GIDArray: An array of XDR-encoded, 32-bit signed integers that contains the group IDs for **UnixAccountName**. The maximum size of this array is MAXGIDS.

2.2.2.20 dump_map_req

This type is used to specify an input parameter to start or continue a map enumeration request to the server. For more details, see sections [2.2.5.5](#), [2.2.5.7](#), [2.2.5.11](#), and [2.2.5.12](#).

```
struct dump map req {
    long PrincipalType;
    long MapRecordIndex;
};
```

PrincipalType: An **XDR**-encoded, 32-bit signed integer that defines the type of account mapping to enumerate. **PrincipalType** MUST be one of the following values.

Value	Meaning
0x00000000	Enumerate user account mappings.
0x00000001	Enumerate group account mappings.

MapRecordIndex: An XDR-encoded, 32-bit signed integer that is an index into the set of mapping records. This MUST be set to 0 on the first call in an enumeration sequence, and to the sum of all the records returned by all preceding replies on subsequent calls in the enumeration sequence. For more information on enumeration sequences, see sections [3.1.5](#) and [3.2.5](#).

2.2.2.21 sequence_number

This type is used by the server to define a version for a set of account mappings at a given point in time. This number is changed by the server whenever any changes are made to the set of account mappings that it maintains (for more details, see section [2.2.5.6](#)). If either of the member fields changes, the sequence_number as a whole MUST be considered as changed.

```
struct sequence_number {
    long CurrentVersionTokenLowPart;
    long CurrentVersionTokenHighPart;
};
```

```
};
```

CurrentVersionTokenLowPart: An **XDR**-encoded, 32-bit signed integer that MUST be either 0x00000000 or a value returned by the server from a previous call to `GETCURRENTVERSIONTOKEN_PROC` or **DUMPALLMAPSXXX_PROC**. For more information about **CurrentVersionTokenLowPart**, see sections [3.1.5](#) and [3.2.5](#).

CurrentVersionTokenHighPart: An XDR-encoded, 32-bit signed integer that MUST be either 0x00000000 or a value returned by the server from a previous call to `GETCURRENTVERSIONTOKEN_PROC` or **DUMPALLMAPSXXX_PROC**. For more information about **CurrentVersionTokenHighPart**, see sections [3.1.5](#) and [3.2.5](#).

2.2.2.22 mapping_record

This type is used to define a single account map when returned as an output from the map enumeration procedure. For more details, see section [2.2.3.1](#).

```
struct mapping_record {
    MapSvrMBCSNameString WindowsAccountName;
    MapSvrMBCSNameString UnixAccountName;
    long ID;
};
```

WindowsAccountName: A [MapSvrMBCSNameString](#) ([section 2.2.2.2](#)) that contains the name of the Windows user or group account in the enumeration. The length of the string MUST NOT exceed 128 bytes. The Windows account name MUST be in the "DOMAIN\NAME" format.

UnixAccountName: A [MapSvrMBCSNameString](#) that contains the name of the **UNIX** user or group account in the enumeration. The length of the string MUST NOT exceed 128 bytes.

ID: An **XDR**-encoded, 32-bit signed integer that contains the UNIX **user ID** or group ID for **UnixAccountName** as specified by **PrincipalType** in the request ([section 2.2.5.5](#)).

2.2.2.23 sid

This type is used to define a Windows account **SID** when used as input to look up the **UNIX** account mapping details that correspond to the Windows account represented by this SID. For more details, see section [2.2.5.10](#) and section [2.2.5.18](#).

```
struct sid {
    char SID<MAXSIDLEN>;
};
```

SID: An array of **XDR**-encoded unsigned bytes that is a stream representation of the Windows account **SID**, as specified in [\[MS-DTYP\]](#) section 2.4.2. The **SubAuthority** field of the **SID** packet ([\[MS-DTYP\]](#) section 2.4.2.2) is a variable-length array of unsigned 32-bit **little-endian** integers. The **sid** structure is an opaque data type generated by the Windows security subsystem. It is not converted to any byte-ordered network representation and SHOULD NOT be interpreted by the User Name Mapping Protocol client or server directly; instead, it SHOULD be supplied to the underlying implementation-defined security subsystem. The maximum size of the **SID** array in the **sid** structure is MAXSIDLEN.

Note Because the **SID** is transmitted as a raw array of bytes, the client and server MUST have identical native **SID** representations for user name mapping to succeed. See section [4.9](#) and section [4.17](#) for examples.

2.2.2.24 mapping_recordW

This type is used to define a single account map when returned as output from the map enumeration procedure. For more details, see section [2.2.3.3](#).

```
struct mapping_recordW {
    MapSvrUnicodeNameString WindowsAccountName;
    MapSvrUnicodeNameString UnixAccountName;
    long ID;
};
```

WindowsAccountName: A [MapSvrUnicodeNameString \(section 2.2.2.3\)](#) that contains the name of the Windows user or group account in the enumeration. The length of the string MUST NOT exceed 256 bytes. The account name MUST be in the form "DOMAIN\NAME".

UnixAccountName: A MapSvrUnicodeNameString that contains the name of the **UNIX** user or group account in the enumeration. The length of the string MUST NOT exceed 256 bytes.

ID: An **XDR**-encoded, 32-bit signed integer that contains the UNIX **user ID** or group ID for **UnixAccountName**, as specified by **PrincipalType** in the request (section [2.2.5.11](#)).

2.2.3 Non-XDR-Compliant Data Structures

There are four data structures that cannot be defined using a pure **XDR** definition. Instead they are defined in terms of lower-level XDR primitives. The difference is as follows. XDR defines fixed-size arrays in terms of constants only. On the other hand, the User Name Mapping Protocol has four structures that use a dynamic value for the array size, and the layout of the fields in the User Name Mapping Protocol precludes the use of the XDR variable-sized array data type. For each of the four data types that follow, the structures are described as their standard XDR types, followed by an XDR vector that uses a dynamic size rather than a constant. This is very similar to the standard XDR variable-sized array but with a separate size value rather than one built into the array type.

See section [7](#) for sample code that shows how to encode and decode each of the four data structures.

2.2.3.1 mapping

This type is used to define a set of account maps when returned as output from the map enumeration procedure. For more details, see section [2.2.5.5](#).

```
struct mapping {
    sequence number Token;
    long MappingRecordCount;
    long TotalMappingRecordCount;
    mapping_record MapArray[MappingRecordCount];
};
```

Token: A [sequence number \(section 2.2.2.21\)](#) that represents the current version of the data set maintained by the User Name Mapping Protocol server.

MappingRecordCount: An **XDR**-encoded, 32-bit signed integer that indicates the number of records that are returned in **MapArray**.

TotalMappingRecordCount: A 32-bit signed integer value that indicates the total number of mapping records of the specified **PrincipalType** (as specified in section 2.2.5.5) held by the server that are available to be enumerated.

MapArray: An array of account mapping records that is returned as a part of the current enumeration sequence, as specified in section [2.2.2.22](#).

2.2.3.2 maps

This type is used to define a set of account maps in colon-delimited string format when returned as output from the map enumeration procedure. For more details, see section [2.2.5.7](#).

```
struct maps {
    sequence_number Token;
    long MappingRecordCount;
    long TotalMappingRecordCount;
    MapSvrMBCSMapString MapArray[MappingRecordCount];
};
```

Token: A sequence of numbers that represent the version for the set of account maps returned in the current enumeration.

MappingRecordCount: An **XDR**-encoded, 32-bit signed integer that indicates the maximum number of records to be returned in the **MapArray** field.

TotalMappingRecordCount: A 32-bit signed integer value that indicates the total number of mapping records of the specified **PrincipalType** (as specified in section 2.2.5.7) held by the server that are available to be enumerated.

MapArray: An array of account mapping records that is returned as a part of the current enumeration sequence (as specified in section [2.2.2.6](#)).

2.2.3.3 mappingW

This type is used to define a set of account maps when returned as output from the map enumeration procedure. For more details, see section [2.2.5.11](#).

```
struct mappingW {
    sequence_number Token;
    long MappingRecordCount;
    long TotalMappingRecordCount;
    mapping_recordW MapArray[MappingRecordCount];
};
```

Token: A sequence number that represents the version for the set of account mappings that are returned in the current enumeration.

MappingRecordCount: An **XDR**-encoded, 32-bit signed integer that indicates the number of records that are returned in the **MapArray** field.

TotalMappingRecordCount: A 32-bit signed integer value that indicates the total number of mapping records of the specified **PrincipalType** (section 2.2.5.11) held by the server that are available to be enumerated.

MapArray: An array of account mapping records that is returned as a part of the current enumeration sequence. For more details, see section [2.2.2.24](#).

2.2.3.4 mapsW

This type is used to define a set of account maps in colon-delimited string format when returned as an output from the map enumeration procedure. For more details, see section [2.2.5.12](#).

```

struct mapsW {
    sequence_number Token;
    long MappingRecordCount;
    long TotalMappingRecordCount;
    MapSvrUnicodeMapString MapArray[MappingRecordCount];
};

```

Token: A [sequence number \(section 2.2.2.21\)](#).

MappingRecordCount: An **XDR**-encoded, 32-bit signed integer that indicates the maximum number of records in **MapArray**.

TotalMappingRecordCount: A 32-bit signed integer value that indicates the total number of mapping records of the specified **PrincipalType** (section 2.2.5.12) held by the server that are available to be enumerated.

MapArray: An array of [MapSvrUnicodeMapString \(section 2.2.2.7\)](#).

2.2.4 Standard Failure Responses

SUNRPC defines a set of standard responses to requests that the User Name Mapping Protocol server is unable to service. The following tables list the set of status codes that can be returned by the User Name Mapping Protocol server.

If SUNRPC status is MSG_ACCEPTED.

Accept status
SUCCESS
PROG_UNAVAIL
PROG_MISMATCH
PROC_UNAVAIL
GARBAGE_ARGS
SYSTEM_ERR

If SUNRPC status is MSG_DENIED.

Reject status	Reason rejected
RPC_MISMATCH	
AUTH_ERROR	AUTH_BADCRED

These status codes have the following meanings:

- SUCCESS: RPC call executed successfully ([\[RFC1057\]](#)).
- PROG_UNAVAIL: Wrong PROGRAM_NUMBER for the port ([RFC1057]).
- PROG_MISMATCH: Unsupported protocol version number requested ([RFC1057]).
- PROC_UNAVAIL: Nonexistent procedure number requested ([RFC1057]).
- GARBAGE_ARGS: Supplied arguments illegal or otherwise not decodable ([RFC1057]).

- SYSTEM_ERR: Errors like memory allocation failure ([\[RFC1831\]](#)).
- RPC_MISMATCH: Invalid SUNRPC version number ([RFC1057]).
- AUTH_ERROR: Remote cannot authenticate caller ([RFC1057]).
- AUTH_BADCRED: Bad credentials in RPC call ([RFC1057]).[<3>](#)

2.2.5 User Name Mapping Protocol Messages

The User Name Mapping Protocol procedure messages are defined in the **SUNRPC** request and response headers, as specified in [\[RFC1057\]](#) section 8. The following table lists the procedure messages in procedure number order.

Procedure name	Procedure number	Version
MAPPROC_NULL	0	1, 2
GETWINDOWSCREDSFROMUNIXUSERNAME_PROC	1	1, 2
GETUNIXCREDSFROMNTUSERNAME_PROC	2	1, 2
AUTHUSINGUNIXCREDS_PROC	3	1, 2
DUMPALLMAPS_PROC	4	1, 2
GETCURRENTVERSIONTOKEN_PROC	5	1, 2
DUMPALLMAPSEX_PROC	6	1, 2
GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC	7	1, 2
GETUNIXCREDSFROMNTGROUPNAME_PROC	8	1, 2
GETUNIXCREDSFROMNTUSERSID_PROC	9	2
DUMPALLMAPSW_PROC	10	2
DUMPALLMAPSEXW_PROC	11	2
GETWINDOWSDUSERFROMUNIXUSERNAMEW_PROC	12	2
GETUNIXCREDSFROMNTUSERNAMEW_PROC	13	2
AUTHUSINGUNIXCREDSW_PROC	14	2
GETWINDOWSGROUPFROMUNIXGROUPNAMEW_PROC	15	2
GETUNIXCREDSFROMNTGROUPNAMEW_PROC	16	2
GETUNIXCREDSFROMNTUSERSIDW_PROC	17	2

2.2.5.1 MAPPROC_NULL (PROC 0)

A null procedure that is used for service discovery as specified in [\[RFC1057\]](#) section A.2.

```
void
MAPPROC_NULL (
void
```

);

This procedure requires no arguments, and a successful reply MUST contain no data other than a **SUNRPC** reply status of MSG_ACCEPTED, as specified in [RFC1057].

The typical use of a null procedure is for the clients to discover whether the service is started and available. This procedure has a procedure number equal to 0.

2.2.5.2 GETWINDOWSCREDSFROMUNIXUSERNAME_PROC (PROC 1)

A request to fetch the mapped Windows user account name for a specified **UNIX** user name and/or UNIX user.

```
windows_creds
GETWINDOWSCREDSFROMUNIXUSERNAME_PROC (
    unix_account UnixUser
);
```

UnixUser: The UNIX user to search for, using the account name and/or UID as the search criteria, as specified by the value for SearchOption.

Return Value: A [windows_creds](#) record that contains the mapped Windows user account details for the specified UNIX user. Whenever the lookup request for a specified UNIX account succeeds or fails to find a corresponding Windows account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. The actual success or failure of the request MUST be set in the Status member of the returned structure. Status is a Boolean value, with 0 indicating a successful lookup request and 1 indicating a failed lookup request.

2.2.5.3 GETUNIXCREDSFROMNTUSERNAME_PROC (PROC 2)

A request to fetch the mapped **UNIX** user account details for a specified Windows user account name.

```
unix_creds
GETUNIXCREDSFROMNTUSERNAME_PROC (
    windows_account WindowsUserAccountName
);
```

WindowsUserAccountName: The Windows user to use for the account name as the search criterion.

Return Value: A [unix_creds](#) record containing the mapped UNIX user account details for the specified Windows account name. Whenever the lookup request for a specified Windows account fails to find a corresponding UNIX account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a zero-length string in the **UnixAccountName** member of the returned structure.

2.2.5.4 AUTHUSINGUNIXCREDS_PROC (PROC 3)

A request to fetch the **UNIX** account details for a given UNIX user name and password.

This procedure is typically used by clients that are doing a simple authentication by providing a user name and password. The password string is the string returned by a call to the crypt function API using the user's cleartext password, as specified in [\[IEEE1003.1\]](#) System Interfaces Volume (XSH) Chapter 3.

```

unix_auth
AUTHUSINGUNIXCREDS_PROC(
    unix_user_auth UnixUserAuth
);

```

UnixUserAuth: UNIX user name and password to use as the search criteria. [<4>](#)

Return Value: A [unix_auth](#) record that contains the mapped UNIX user account details for the specified UNIX account. Whenever the lookup request for a specified UNIX account fails to find a corresponding UNIX account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a zero-length string in the **UnixAccountPassword** member of the returned structure.

2.2.5.5 DUMPALLMAPS_PROC (PROC 4)

A request to enumerate all account mappings held by the service.

```

mapping
DUMPALLMAPS_PROC(
    dump_map_req EnumCursor
);

```

EnumCursor: A **PrincipalType** and index to start or continue an enumeration.

Return Value: A mapping type that describes an array of zero or more [mapping_record](#) types. Whenever the enumeration request fails to find any records to either begin or continue the enumeration, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS, and MUST return 0 in the **MappingRecordCount** field. It MUST also return a zero-length set of [mapping_record](#) types in the **MapArray** member of the returned structure. The User Name Mapping Protocol server MUST also return current values for the server [sequence_number](#) in the **Token** field and the total mapping record count for the specified enumeration in the **TotalMappingRecordCount** field of the returned structure.

2.2.5.6 GETCURRENTVERSIONTOKEN_PROC (PROC 5)

A request for the current account-mapping sequence number for the set of mapping records held by the server. This procedure is used by clients to check whether any **map** records changed since the last enumeration by the client.

```

sequence_number
GETCURRENTVERSIONTOKEN_PROC(
    sequence_number SequenceNumber
);

```

SequenceNumber: A data structure that contains two 32-bit signed integers. *SequenceNumber* MUST contain either 0x00000000 for each member field or a value returned by the server from a previous call to GETCURRENTVERSIONTOKEN_PROC or **DUMPALLMAPSXXX_PROC**. For more details, see sections [3.1.5](#) and [3.2.5](#).

Return Value: The User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a [sequence_number](#) structure with the current sequence number value for the set of mapping records held by the server.

2.2.5.7 DUMPALLMAPSEX_PROC (PROC 6)

A request to enumerate all account mappings held by the service.

```
maps
DUMPALLMAPSEX_PROC (
    dump_map_req EnumCursor
);
```

EnumCursor: A **PrincipalType** and index to start or continue an enumeration.

Return Value: A [maps](#) type record that describes an array of zero or more [MapSvrMBCSMapString](#) types. Whenever the enumeration request fails to find any records to either begin or continue the enumeration, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS, and MUST return 0 in the **MappingRecordCount** field. It MUST also return a zero-length set of **MapSvrMBCSMapString** types in the **MapArray** member of the returned structure.

The User Name Mapping Protocol server MUST also return current values for the server [sequence number](#) in the **Token** field and the total mapping record count for the specified enumeration in the **TotalMappingRecordCount** field of the returned structure.

2.2.5.8 GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC (PROC 7)

A request to fetch the Windows group account information that corresponds to a **UNIX** group name.

```
windows_creds
GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC (
    unix_account UnixGroupAccount
);
```

UnixGroupAccount: A UNIX group to search for, using the account name and/or **GID** as the search criteria, as specified by the value for **SearchOption**.

Return Value: A [windows_creds](#) record that contains the mapped Windows group account details for the specified UNIX group. Whenever the lookup request for a specified UNIX account succeeds or fails to find a corresponding Windows account **map**, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. The actual success or failure of the request MUST be set in the **Status** member of the returned structure. **Status** is a Boolean value, with 0 indicating a successful lookup request and 1 indicating a failed lookup request.

2.2.5.9 GETUNIXCREDSFROMNTGROUPNAME_PROC (PROC 8)

A request to fetch the **UNIX** group account information that corresponds to a Windows group name.

```
unix_creds
GETUNIXCREDSFROMNTGROUPNAME_PROC (
    windows_account WindowsGroupAccountName
);
```

WindowsGroupAccountName: A Windows group to use for the account name as the search criteria.

Return Value: A [unix_creds](#) record that contains the mapped UNIX group account details for the specified Windows account name. Whenever the lookup request for a specified Windows account fails to find a corresponding UNIX account **map**, the User Name Mapping Protocol server MUST

return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a zero-length string in the **UnixAccountName** member of the returned structure.

2.2.5.10 GETUNIXCREDSFROMNTUSERSID_PROC (PROC 9)

A request for the **UNIX** account information that corresponds to the Windows account specified by the **SID**.

```
unix creds
GETUNIXCREDSFROMNTUSERSID_PROC (
    sid SID
);
```

SID: A Windows SID to use as the search criteria.

Return Value: A [unix creds](#) record that contains the mapped UNIX account details for the specified Windows SID. Whenever the lookup request for a specified Windows SID fails to find a corresponding UNIX account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a zero-length string in the **UnixAccountName** member of the returned structure.

2.2.5.11 DUMPALLMAPSW_PROC (PROC 10)

This procedure is the **wide character** counterpart of [DUMPALLMAPS_PROC](#). The request and response packets are identical to **DUMPALLMAPS_PROC**, except that the return value is a [mappingW](#) data type instead of a [mapping](#) data type. For example, the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
mappingW
DUMPALLMAPSW_PROC (
    dump map_req EnumCursor
);
```

EnumCursor: A **PrincipalType** and index to start or continue an enumeration.

Return Value: A mappingW type that describes an array of zero or more [mapping_recordW](#) types. Whenever the enumeration request fails to find any records to either begin or continue the enumeration, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS, and MUST return 0 in the **MappingRecordCount** field. It MUST also return a zero-length set of mapping_recordW types in the **MapArray** member of the returned structure. The User Name Mapping Protocol server MUST also return current values for the server [sequence number](#) in the **Token** field, and the total mapping record count for the specified enumeration in the **TotalMappingRecordCount** field of the returned structure.

2.2.5.12 DUMPALLMAPSEXW_PROC (PROC 11)

This procedure is the **wide character** counterpart of [DUMPALLMAPSEX_PROC](#). The request and response packets are identical to **DUMPALLMAPSEX_PROC**, except that the [MapSvrMBCSMapString](#) data type is replaced with a [MapSvrUnicodeMapString](#) type in the byte stream.

```
mapsW
DUMPALLMAPSEXW_PROC (
    dump_map_req EnumCursor
);
```

EnumCursor: A **PrincipalType** and index to start or continue an enumeration.

Return Value: A [mapsW](#) type record that describes an array of zero or more MapSvrUnicodeMapString types. Whenever the enumeration request fails to find any records to either begin or continue the enumeration, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS, and MUST return 0 in the **MappingRecordCount** field. It MUST also return a zero-length set of **MapSvrUnicodeMapString** types in the **MapArray** member of the returned structure. The User Name Mapping Protocol server MUST also return current values for the server [sequence_number](#) in the **Token** field, and the total mapping record count for the specified enumeration in the **TotalMappingRecordCount** field of the returned structure.

2.2.5.13 GETWINDOWSUSERFROMUNIXUSERNAMEW_PROC (PROC 12)

This procedure is the **wide character** counterpart to [GETWINDOWSCREDSFROMUNIXUSERNAME_PROC](#). The request and response packets are identical to **GETWINDOWSCREDSFROMUNIXUSERNAME_PROC**, except that the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
windows_credsW
GETWINDOWSUSERFROMUNIXUSERNAMEW_PROC(
    unix_accountW UnixUser
);
```

UnixUser: A **UNIX** account to search for, using the account name and/or UID as the search criteria, as specified by the value for **SearchOption**.

Return Value: A [windows_credsW](#) record that contains the mapped Windows user account details for the specified UNIX user account. Whenever the lookup request for a specified UNIX account succeeds or fails to find a corresponding Windows account **map**, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. The actual success or failure of the request MUST be set in the **Status** member of the returned structure. **Status** is a Boolean value, with 0 indicating a successful lookup request and 1 indicating a failed lookup request.

2.2.5.14 GETUNIXCREDSFROMNTUSERNAMEW_PROC (PROC 13)

This procedure is the **wide character** counterpart to [GETUNIXCREDSFROMNTUSERNAME_PROC](#). The request and response packets are identical to **GETUNIXCREDSFROMNTUSERNAME_PROC**, except that the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
unix_credsW
GETUNIXCREDSFROMNTUSERNAMEW_PROC(
    windows_accountW WindowsUserAccountName
);
```

WindowsUserAccountName: A Windows account to use for the account name as the search criterion.

Return Value: A [unix_credsW](#) record that contains the mapped **UNIX** user account details for the specified Windows account name. Whenever the lookup request for a specified Windows account fails to find a corresponding UNIX account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a zero-length string in the **UnixAccountName** member of the returned structure.

2.2.5.15 AUTHUSINGUNIXCREDSW_PROC (PROC 14)

This procedure is the **wide character** counterpart to [AUTHUSINGUNIXCREDS_PROC](#). The request and response packets are identical to **AUTHUSINGUNIXCREDS_PROC**, except that the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
unix_authW
AUTHUSINGUNIXCREDSW_PROC(
    unix_user_authW UnixUserAuth
);
```

UnixUserAuth: A **UNIX** user name and password to use as the search criteria. [<5>](#)

Return Value: A [unix_authW](#) record that contains the mapped UNIX user account details for the specified UNIX account. Whenever the lookup request for a specified UNIX account fails to find a corresponding UNIX account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It MUST also return a zero-length string in the **UnixAccountPassword** member of the returned structure.

2.2.5.16 GETWINDOWSGROUPFROMUNIXGROUPNAMEW_PROC (PROC 15)

This procedure is the **wide character** counterpart to [GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC](#). The request and response packets are identical to **GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC**, except that the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
windows_credsW
GETWINDOWSGROUPFROMUNIXGROUPNAMEW_PROC(
    unix_accountW UnixGroupAccount
);
```

UnixGroupAccount: A **UNIX** group to search for, using the account name and/or **GID** as the search criteria, as specified by the value for **SearchOption**.

Return Value: A [windows_credsW](#) record that contains the mapped Windows group account details for the specified UNIX group. Whenever the lookup request for a specified UNIX account succeeds or fails to find a corresponding Windows account map, the User Name Mapping Protocol server MUST return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. The actual success or failure of the request MUST be set in the **Status** member of the returned structure. **Status** is a Boolean value, with 0 indicating a successful lookup request and 1 indicating a failed lookup request.

2.2.5.17 GETUNIXCREDSFROMNTGROUPNAMEW_PROC (PROC 16)

This procedure is the **wide character** counterpart to [GETUNIXCREDSFROMNTGROUPNAME_PROC](#). The request and response packets are identical to **GETUNIXCREDSFROMNTGROUPNAME_PROC**, except that the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
unix_credsW
GETUNIXCREDSFROMNTGROUPNAMEW_PROC(
    windows_accountW WindowsGroupAccountName
);
```

WindowsGroupName: A Windows group to use as the account name in the search criteria.

Return Value: A [unix_credsW](#) record that contains the mapped **UNIX** group account details for the specified Windows account name. Whenever the lookup request for a specified Windows account fails to find a corresponding UNIX account map, the User Name Mapping Protocol server **MUST** return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It **MUST** also return a zero-length string in the **UnixAccountName** member of the returned structure.

2.2.5.18 GETUNIXCREDSFROMNTUSERSIDW_PROC (PROC 17)

This procedure is the **wide character** counterpart to [GETUNIXCREDSFROMNTUSERSID_PROC](#). The request and response packets are identical to **GETUNIXCREDSFROMNTUSERSID_PROC**, except that the [MapSvrMBCSNameString](#) data type is replaced with a [MapSvrUnicodeNameString](#) type in the byte stream.

```
unix_credsW
GETUNIXCREDSFROMNTUSERSIDW PROC (
    sid SID
);
```

SID: A Windows **SID** to use as the search criteria.

Return Value: A [unix_credsW](#) record that contains the mapped **UNIX** account details for the specified Windows SID. Whenever the lookup request for a specified Windows SID fails to find a corresponding UNIX account map, the User Name Mapping Protocol server **MUST** return a **SUNRPC** status of MSG_ACCEPTED with an accept status of SUCCESS. It **MUST** also return a zero-length string in the **UnixAccountName** member of the returned structure.

3 Protocol Details

With the exception of the **DUMPALLMAPSXXX_PROC** procedures, requests sent by the User Name Mapping Protocol client generate a single response from the User Name Mapping Protocol server. There is no predetermined sequencing.

The **DUMPALLMAPSXXX_PROC** procedures are used to enumerate some, or all the mapping records held by the User Name Mapping Protocol server. All the **DUMPALLMAPSXXX_PROC** procedures follow the same sequencing rules, as defined in the following sections. The sequence can be restricted to a single request-response pair, or it can extend over many request-response pairs, depending on the number of maps available on the User Name Mapping Protocol server and the requirements of the User Name Mapping Protocol client.

Each enumeration sequence is independent of other individual requests or enumeration sequences between the User Name Mapping Protocol client and server. Therefore, multiple enumerations (from the same or different clients) for **user map** and **group map** can proceed in parallel without any interference.

3.1 Client Details

3.1.1 Abstract Data Model

This section describes a conceptual model of possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model, as long as their external behavior is consistent with that described in this document.

Clients of the User Name Mapping Protocol can maintain copies of user mappings (and group mappings) enumerated from the server. Clients can use the **DUMPALLMAPSXXX_PROC** procedures to enumerate all individual maps from the server. The server treats account mappings as an unordered array of mapping records of total count equal to **TotalMappingRecordCount**, as explained in sections [2.2.3.1](#) and [2.2.3.3](#). The index of records begins at zero, and **MappingRecordCount** indicates the number of **map** records returned by the server in the current RPC response packet.

Clients can cache **CurrentVersionTokenHighPart** and **CurrentVersionTokenLowPart** values returned by the **DUMPALLMAPSXXX_PROC** response to implement cache consistency. Cache consistency is implemented on clients by periodically polling the server's [GETCURRENTVERSIONTOKEN_PROC](#) procedure to know when to refresh their locally cached copies of mappings.

As an alternative to the enumeration request (**DUMPALLMAPSXXX_PROC**), the clients can cache the results of individual account lookup requests and use **GETCURRENTVERSIONTOKEN_PROC** to know when to refresh their locally cached copies of mappings.

Clients of the User Name Mapping Protocol are at liberty to implement caching and persistence in any way they please. The User Name Mapping Protocol server functions as a read-only lookup service of account mappings.

The following figure shows the data model for a client of the User Name Mapping Protocol.

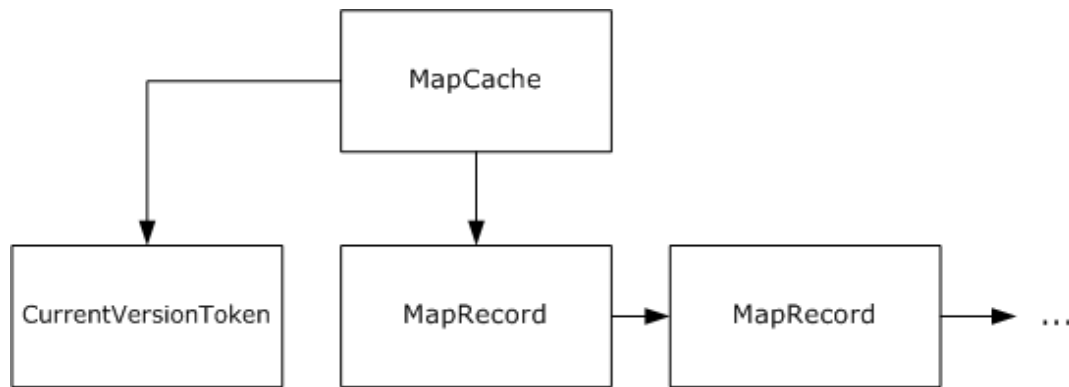


Figure 1: User Name Mapping Protocol data model: Client

There are three elements in the model: MapCache, MapRecord, and CurrentVersionToken.

MapCache: The MapCache element models the information that the client has collected from the server by enumerating maps using the DUMPALLMAPSXXX_PROC. The MapCache element contains a list (or array) of MapRecord elements, each of which describes the mapping between a Windows and a **UNIX** account.

MapRecord: The MapRecord element models the information for a single Windows-to-UNIX user account mapping or group account mapping. It contains the UNIX account name and UID, a **GID**, and the supplementary GID details that correspond to a Windows account name and **domain**.

CurrentVersionToken: This element models the version of the cache as a whole. This element is guaranteed by the server to be different for different versions of the MapCache. Clients can use this element to implement cache consistency with respect to the server by periodically polling this token by using the GETCURRENTVERSIONTOKEN_PROC procedure.

3.1.2 Timers

There are no timers in the User Name Mapping Protocol beyond those used by **SUNRPC**.

3.1.3 Initialization

None.

3.1.4 Higher-Layer Triggered Events

None.

3.1.5 Message Processing Events and Sequencing Rules

The User Name Mapping Protocol allows a User Name Mapping Protocol client to retrieve a complete set of account mappings from the server and to maintain a copy of these mappings in a local cache. The client uses a combination of the **DUMPALLMAPSXXX_PROC** and [GETCURRENTVERSIONTOKEN_PROC](#) procedure calls to retrieve the account mappings and to check for updates to the account mappings in the server, respectively. The **DUMPALLMAPSXXX_PROC** procedure that is chosen is determined by the type of information that the User Name Mapping Protocol client chooses to cache.

All procedures other than **DUMPALLMAPSXXX_PROC** are self-contained in that they do not require any other procedures to be sequenced to complete successfully. The User Name Mapping Protocol client does not need to maintain any state to implement sequencing across procedure calls.

For all procedures, the processing rules for a server-returned response packet are specified in [\[RFC1057\]](#) section 8. The client MUST interpret server procedure response status of MSG_ACCEPTED or MSG_DENIED according to those rules.

3.1.5.1 Making the Initial Account Mapping Request to the Server

The sequence begins when a User Name Mapping Protocol client sends a **DUMPALLMAPSXXX_PROC** procedure request to the server, with the **MapRecordIndex** field equal to 0 to indicate the start of a new enumeration sequence and the **PrincipalType** field equal to the record type to be returned.

3.1.5.2 Processing the Account Mapping Response from the Server

If the **DUMPALLMAPSXXX_PROC** response from the server indicates success and the returned value of **MappingRecordCount** is less than the returned value of **TotalMappingRecordCount**, the client proceeds to section [3.1.5.3](#); the enumeration of account mappings returned from the server is incomplete and there are more records to retrieve.

Otherwise, the enumeration returned is complete if the response indicates success. The client MAY send another **DUMPALLMAPSXXX_PROC** request to the server if the response indicates failure.

3.1.5.3 Making Further Account Mapping Requests to the Server

The User Name Mapping Protocol client continues to make further **DUMPALLMAPSXXX_PROC** requests, each time increasing the value of **MapRecordIndex** to the total number of map records returned by the server so far for this enumeration. For example, if the first reply returned 15 records, and the second reply returned 12 records, the third request in the sequence sets the **MapRecordIndex** to 27 (15 + 12). The User Name Mapping Protocol client continues to make requests until there are no more account mappings to retrieve from the server. This is indicated by a **DUMPALLMAPSXXX_PROC** reply that contains zero records (**MappingRecordCount** is 0), or if the next **DUMPALLMAPSXXX_PROC** request would set **MapRecordIndex** to **TotalMappingRecordCount**. **TotalMappingRecordCount** is returned in the server **DUMPALLMAPSXXX_PROC** response.

If at any point the values of **CurrentVersionTokenHighPart**, **CurrentVersionTokenLowPart**, or **TotalMappingRecordCount** returned by the server in the **DUMPALLMAPSXXX_PROC** response change from the initial values returned when **MapRecordIndex** was set to 0 in the **DUMPALLMAPSXXX_PROC** request, the current enumeration MUST be abandoned and restarted with a new **DUMPALLMAPSXXX_PROC** request (**MapRecordIndex** equal to 0).

3.1.5.4 Polling for Cache Consistency

The User Name Mapping Protocol client uses [GETCURRENTVERSIONTOKEN_PROC](#) to periodically check the server for cache consistency. Whenever any of the user or group account mappings on the server change, the tokens returned in the response to **GETCURRENTVERSIONTOKEN_PROC** are different, at which point the client MUST discard its cached copy of all the mappings in their entirety and enumerate the new set of mappings from the server.

If **CurrentVersionTokenHighPart** and **CurrentVersionTokenLowPart** in the **GETCURRENTVERSIONTOKEN_PROC** reply are the same as those from the previous enumeration, there have been no changes to any map records, and any cache of map records being maintained by the User Name Mapping Protocol client is still valid. If either **CurrentVersionTokenHighPart** or **CurrentVersionTokenLowPart** in the **GETCURRENTVERSIONTOKEN_PROC** reply differs from those returned by the previous enumeration, the mapping records have been updated, and the User Name Mapping Protocol client MUST consider the local cached copies of the mapping records as out of date and MUST repeat the enumeration to get the updated set of mapping records.

3.1.6 Timer Events

None.

3.1.7 Local Events

None.

3.2 Server Details

3.2.1 Abstract Data Model

This section describes a conceptual model of possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model, as long as their external behavior is consistent with that described in this document.

The User Name Mapping Protocol server maintains a database of account mappings and provides procedures for enumeration of these account mappings. The server maintains a unique 64-bit sequence number that is initialized at server startup and changed whenever the database of maps is updated.

The User Name Mapping Protocol server returns the 64-bit sequence number to the clients to allow them to implement a polling-based cache consistency scheme that times out locally cached copies of account mappings on the client.

The following figure shows the data model for the User Name Mapping Protocol server.

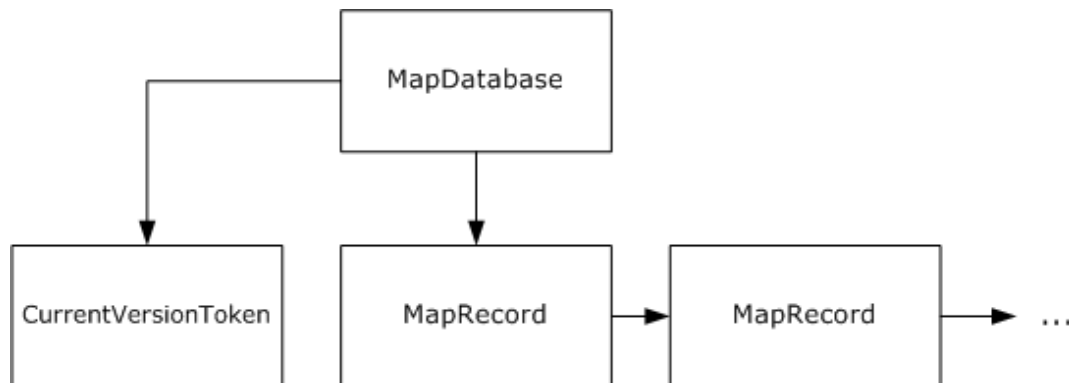


Figure 2: User Name Mapping Protocol data model: Server

There are three elements in the model: MapDatabase, MapRecord, and CurrentVersionToken.

MapDatabase: The MapDatabase element models a nonvolatile store of mapping information between Windows and **UNIX** accounts. This element contains a set of MapRecord elements and a CurrentVersionToken element.

MapRecord: The MapRecord element models the information for a single Windows-to-UNIX user account mapping or group account mapping. It contains the UNIX account name and UID, a **GID**, and the supplementary GID details that correspond to a Windows account name and **domain**.

CurrentVersionToken: This element models the version of the MapDatabase as a whole. This element **MUST** be guaranteed by the server to be unique following each update to the MapDatabase.

3.2.2 Timers

None.

3.2.3 Initialization

None.

3.2.4 Higher-Layer Triggered Events

None.

3.2.5 Message Processing Events and Sequencing Rules

3.2.5.1 Processing for All Procedures

The User Name Mapping Protocol server performs a simple lookup or enumeration service on behalf of clients. As defined in section [3.2.1](#), the server maintains a set of current mappings that it traverses to answer queries by clients. For each lookup procedure from the client, the User Name Mapping Protocol server queries the persistent data store of account mappings and returns details of the located map, if found.

The **SUNRPC** response packet generated by the User Name Mapping Protocol server adheres to the rules indicated in [\[RFC1057\]](#) section 8. Whenever a well-formed SUNRPC request is received, the body of the response packet **MUST** have a status of MSG_ACCEPTED to indicate a successful receipt of the packet.[<6>](#)

The server **MUST** return an error of SUNRPC PROG_MISMATCH whenever the client requests a program version other than 1 or 2.

In all cases where the server fails to decode the lookup or enumeration procedure request arguments, it **MUST** return a response error value of GARBAGE_ARGS.

In all cases where the lookup or enumeration request succeeds, the server **MUST** return a SUCCESS status in the reply body and encode the procedure-specific return values according to the **XDR** rules defined in [\[RFC4506\]](#).

3.2.5.2 Processing of DUMPALLMAPSXXX_PROC Request and GETCURRENTVERSIONTOKEN_PROC Request

3.2.5.2.1 Processing the Initial Account Mapping Request from the Client

The User Name Mapping Protocol server replies to the **DUMPALLMAPSXXX_PROC** request with a two-part version token (**CurrentVersionTokenHighPart** and **CurrentVersionTokenLowPart**), a count of the number of maps in the reply (**MappingRecordCount**), the total number of maps available on the server (**TotalMappingRecordCount**), and a list of **MappingRecordCount** mapping records that begin at the **MapRecordIndex** index equal to 0. The number of account mapping records returned by the server to the client is implementation specific.[<7><8>](#)

3.2.5.2.2 Processing Further Account Mapping Requests from the Client

The User Name Mapping Protocol server replies to the **DUMPALLMAPSXXX_PROC** request with the next set of mapping records, starting with the map record at the **MapRecordIndex** index requested. If the **MapRecordIndex** requested is out of bounds of the **TotalMappingRecordCount** number of account mappings stored on the server, **MappingRecordCount** **MUST** be returned with a value of 0, and no records are returned. The number of account mapping records returned by the server to the

client is implementation specific. For example, the server might limit the number of mappings returned to the amount of data that can fit in a single **SUNRPC** packet of a chosen maximum size. <9><10><11>

3.2.5.2.3 Processing the Client Account Mapping Cache Refresh

The User Name Mapping Protocol server replies with **CurrentVersionTokenHighPart** and **CurrentVersionTokenLowPart** in the [GETCURRENTVERSIONTOKEN_PROC](#) reply set to an implementation-specific value. If the account mappings have been changed since a client's previous **GETCURRENTVERSIONTOKEN_PROC** or **DUMPALLMAPSXXX_PROC** enumeration request, the values returned to the client MUST be different from the values returned for the previous request to **GETCURRENTVERSIONTOKEN_PROC** or **DUMPALLMAPSXXX_PROC**. (The method used to track changes in account mappings is implementation specific.) If the account mappings have not changed, the values returned to the client MUST be the values returned for the previous request to **GETCURRENTVERSIONTOKEN_PROC** or **DUMPALLMAPSXXX_PROC**. <12>

3.2.6 Timer Events

None.

3.2.7 Other Local Events

None.

4 Protocol Examples

Several examples of network traffic for common User Name Mapping Protocol **SUNRPC** procedures are outlined in the following sections, giving an indication of normal traffic flow. The example network traffic is illustrated with the aid of the following sample user and group mapping database at the server. In this example, a sample User Name Mapping Protocol SUNRPC service has been configured on the server to map users in the Windows domain "nfs-dom-1" to **POSIX** user and **group identifiers**.

Advanced **User Mappings**

Windows user	POSIX user	UID	GID
nfs-dom-1\administrator	Root	0	0
nfs-dom-1\u1	u1	401	401
nfs-dom-1\u2	u2	402	401
nfs-dom-1\u3	u3	403	402

Advanced **Group Mappings**

Windows group	POSIX group	GID
nfs-dom-1\Domain Admins	bin	1
nfs-dom-1\g1	g1	401
nfs-dom-1\g2	g3	402

Simple User Mappings

Windows user	POSIX user	UID	GID
nfs-dom-1\spec	spec	500	500
nfs-dom-1\u4	u4	404	402
nfs-dom-1\u5	u5	405	401
nfs-dom-1\u6	u6	406	402

Simple Group Mappings

Windows group	POSIX group	GID
nfs-dom-1\specgroup	specgroup	500
nfs-dom-1\g4	g4	404

4.1 GETWINDOWSCREDSFROMUNIXUSERNAME_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the Windows account mapping for **POSIX** user "root". The client asks for a match on the POSIX username alone in the **SearchOption** of the procedure.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 33455, Total IP Length = 88
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 68
- Rpc: Call, Program = mapsvc, Procedure =
  GETWINDOWSCREDSFROMUNIXUSERNAME PROC
  TransactionID: 1221413202 (0x48CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETWINDOWSCREDSFROMUNIXUSERNAME_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: GETWINDOWSCREDSFROMUNIXUSERNAME_PROC Call
- UnixUser:
  SearchOption: UnixAccountName is valid (0x1)
  Reserved: Send as 0x00000000
  ID: 0
- UnixAccountName: 0x1
  - UNMName: root
    Length: 4
    Data: root

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **advanced map** for POSIX user "root" to Windows user "nfs-dom-1\administrator", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 62340, Total IP Length = 88
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 68
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1221413202 (0x48CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
    AcceptState: Call succeeded
- Unm: GETWINDOWSCREDSFROMUNIXUSERNAME PROC Reply
- WindowsCreds:
  Status: 0
  Reserved: Send as 0x00000000
- WindowsAccountName: 0x1
  - UNMWindowsName: nfs-dom-1\administrator
    Length: 23
    Data: nfs-dom-1\administrator
    Padding: Binary Large Object (1 Bytes)

```

4.2 GETUNIXCREDSFROMNTUSERNAME_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service that requests the **POSIX** account mapping for the Windows user "nfs-dom-1\administrator".

```

Frame:

```

```

+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 40198, Total IP Length = 96
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 76
- Rpc: Call, Program = mapsvc, Procedure =
  GETUNIXCREDSFROMNTUSERNAME_PROC
  TransactionID: 1305299282 (0x4DCD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETUNIXCREDSFROMNTUSERNAME_PROC
  - Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: GETUNIXCREDSFROMNTUSERNAME PROC Call
  - WindowsUserAccountName:
    - WindowsAccountName: 0x1
      - UNMName: nfs-dom-1\administrator
        Length: 23
        Data: nfs-dom-1\administrator
        Padding: Binary Large Object (1 Bytes)

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **advanced map** for Windows user "nfs-dom-1\administrator" to POSIX user "root" with UID 0 and **GID 0**, illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 20813, Total IP Length = 76
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 56
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1305299282 (0x4DCD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
  - MessageAccepted:
    - Verification:
      Flavor: No Identity Authentication
      AuthDataLength: 0 (0x0)
      AcceptState: Call succeeded
- Unm: GETUNIXCREDSFROMNTUSERNAME_PROC Reply
  - UnixCreds:
    - UnixAccountName: 0x1
      - UNMName: root
        Length: 4
        Data: root
      ID: 0
      GidCount: 2
    - GID:
      GID: 1
      GID: 1

```

4.3 AUTHUSINGUNIXCREDS_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** account details for the POSIX user "root" with an empty password.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)

```

```

+ Ipv4: Next Protocol = UDP, Packet ID = 41135, Total IP Length = 80
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 60
- Rpc: Call, Program = mapsvc, Procedure = AUTHUSINGUNIXCREDS_PROC
    TransactionID: 1322076498 (0x4ECD4952)
    MessageType: Call
- ServiceCall:
    RPCVersionNumber: 2 (0x2)
    ProgramNumber: mapsvc, 351455(0x00055CDF)
    ProgramVersion: 2 (0x2)
    ProcedureNumber: AUTHUSINGUNIXCREDS_PROC
- Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: AUTHUSINGUNIXCREDS PROC Call
- UnixUserAuth:
- UnixUserAccountName: 0x1
    - UNMName: root
      Length: 4
      Data: root
- UnixUserAccountPassword: 0x1
    - UNMName:
      Length: 0
      Data:

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the mapped POSIX account details for the user "root", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 25985, Total IP Length = 76
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 56
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
    TransactionID: 1322076498 (0x4ECD4952)
    MessageType: Reply
- ServiceReply:
    ReplyStatus: Message accepted
- MessageAccepted:
- Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
    AcceptState: Call succeeded
- Unm: AUTHUSINGUNIXCREDS_PROC Reply
- UnixCreds:
- UnixUserAccountPassword: 0x1
    - UNMName: x
      Length: 1
      Data: x
      Padding: Binary Large Object (3 Bytes)
    ID: 0
    GidCount: 2
- GID:
    GID: 1
    GID: 1

```

4.4 DUMPALLMAPS_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service that requests an enumeration of all user maps (**PrincipalType=0**) starting at index zero (**MapRecordIndex=0**).

```

Frame:

```

```

+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 57181, Total IP Length = 76
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 56
- Rpc: Call, Program = mapsvc, Procedure = DUMPALLMAPS_PROC
  TransactionID: 1238190418 (0x49CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: DUMPALLMAPS_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: DUMPALLMAPS_PROC Call
- EnumCursor:
  PrincipalType: Enumerate user account mappings (0)
  MapRecordIndex: 0

```

The User Name Mapping Protocol service on the server responds with a listing of advanced and simple user mappings in the database. The response packet includes a sequence number that indicates the version for the current set of account mappings, a record count that indicates the number of mappings returned as a part of the current packet payload, the total number of maps in the database of the requested types, and finally, the individual maps themselves.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 45847, Total IP Length = 308
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 288
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1238190418 (0x49CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
  AcceptState: Call succeeded
- Unm: DUMPALLMAPS_PROC Reply
- Mapping:
- Token:
  CurrentVersionTokenLowPart: 19924186
  CurrentVersionTokenHighPart: 0
  MappingRecordCount: 8
  TotalMappingRecordCount: 8
- Map:
- WindowsAccountName: 0x1
  - UNMName: nfs-dom-1\administrator
    Length: 23
    Data: nfs-dom-1\administrator
    Padding: Binary Large Object (1 Bytes)
- UnixAccountName: 0x1
  - UNMName: root
    Length: 4
    Data: root
    ID: 0
- Map:
- WindowsAccountName: 0x1
  - UNMName: NFS-DOM-1\u1
    Length: 12
    Data: NFS-DOM-1\u1
- UnixAccountName: 0x1

```

- UNMName: u1
 - Length: 2
 - Data: u1
 - Padding: Binary Large Object (2 Bytes)
- ID: 401
- Map:
 - WindowsAccountName: 0x1
 - UNMName: NFS-DOM-1\u2
 - Length: 12
 - Data: NFS-DOM-1\u2
 - UnixAccountName: 0x1
 - UNMName: u2
 - Length: 2
 - Data: u2
 - Padding: Binary Large Object (2 Bytes)
- ID: 402
- Map:
 - WindowsAccountName: 0x1
 - UNMName: NFS-DOM-1\u3
 - Length: 12
 - Data: NFS-DOM-1\u3
 - UnixAccountName: 0x1
 - UNMName: u3
 - Length: 2
 - Data: u3
 - Padding: Binary Large Object (2 Bytes)
- ID: 403
- Map:
 - WindowsAccountName: 0x1
 - UNMName: NFS-DOM-1\spec
 - Length: 14
 - Data: NFS-DOM-1\spec
 - Padding: Binary Large Object (2 Bytes)
 - UnixAccountName: 0x1
 - UNMName: spec
 - Length: 4
 - Data: spec
- ID: 500
- Map:
 - WindowsAccountName: 0x1
 - UNMName: NFS-DOM-1\u4
 - Length: 12
 - Data: NFS-DOM-1\u4
 - UnixAccountName: 0x1
 - UNMName: u4
 - Length: 2
 - Data: u4
 - Padding: Binary Large Object (2 Bytes)
- ID: 404
- Map:
 - WindowsAccountName: 0x1
 - UNMName: NFS-DOM-1\u5
 - Length: 12
 - Data: NFS-DOM-1\u5
 - UnixAccountName: 0x1
 - UNMName: u5
 - Length: 2
 - Data: u5
 - Padding: Binary Large Object (2 Bytes)
- ID: 405
- Map:
 - WindowsAccountName: 0x1
 - UNMName: NFS-DOM-1\u6
 - Length: 12
 - Data: NFS-DOM-1\u6
 - UnixAccountName: 0x1
 - UNMName: u6
 - Length: 2
 - Data: u6

Padding: Binary Large Object (2 Bytes)
ID: 406

4.5 GETCURRENTVERSIONTOKEN_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service that requests the current account mapping sequence number for the set of mapping records held by the server.

```
Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 47908, Total IP Length = 76
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 56
- Rpc: Call, Program = mapsvc, Procedure = GETCURRENTVERSIONTOKEN_PROC
  TransactionID: 1422739794 (0x54CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETCURRENTVERSIONTOKEN_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: GETCURRENTVERSIONTOKEN PROC Call
- SequenceNumber:
  CurrentVersionTokenLowPart: 11337900
  CurrentVersionTokenHighPart: 0
```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the current sequence number value for the set of mapping records held by it, illustrated as follows.

```
Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 38726, Total IP Length = 60
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 40
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1422739794 (0x54CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
  AcceptState: Call succeeded
- Unm: GETCURRENTVERSIONTOKEN_PROC Reply
- SequenceNumber:
  CurrentVersionTokenLowPart: 19924186
  CurrentVersionTokenHighPart: 0
```

4.6 DUMPALLMAPSEX_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting an enumeration of all user maps (**PrincipalType=0**) starting at index zero (**MapRecordIndex=0**).

```
Frame:
+ Ethernet: Etype = Internet IP (IPv4)
```

```

+ Ipv4: Next Protocol = UDP, Packet ID = 48740, Total IP Length = 76
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 56
- Rpc: Call, Program = mapsvc, Procedure = DUMPALLMAPSEX_PROC
    TransactionID: 1439517010 (0x55CD4952)
    MessageType: Call
- ServiceCall:
    RPCVersionNumber: 2 (0x2)
    ProgramNumber: mapsvc, 351455 (0x00055CDF)
    ProgramVersion: 2 (0x2)
    ProcedureNumber: DUMPALLMAPSEX_PROC
- Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: DUMPALLMAPSEX PROC Call
- EnumCursor:
    PrincipalType: Enumerate user account mappings
    MapRecordIndex: 0

```

The User Name Mapping Protocol service on the server responds with a listing of advanced and simple user mappings in the database. The response packet includes a sequence number that indicates the version for the current set of account mappings, a record count that indicates the number of mappings returned as a part of the current packet payload, the total number of maps in the database of the requested types, and finally, the individual maps themselves.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 42795, Total IP Length = 464
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 444
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
    TransactionID: 1439517010 (0x55CD4952)
    MessageType: Reply
- ServiceReply:
    ReplyStatus: Message accepted
- MessageAccepted:
- Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
    AcceptState: Call succeeded
- Unm: DUMPALLMAPSEX PROC Reply
- Maps: Count = 8
- Token:
    CurrentVersionTokenLowPart: 19924186
    CurrentVersionTokenHighPart: 0
    MappingRecordCount: 8
    TotalMappingRecordCount: 8
- Map: 0x1
- UNMMapString: *:nfs-dom-1\administrator:0:PCNFS:PCNFS:
    root:x:0:1:1
    Length: 52
    Data: *:nfs-dom-1\administrator:0:PCNFS:PCNFS:root:x:0:1:1
- Map: 0x1
- UNMMapString: *:NFS-DOM-1\u1:0:PCNFS:PCNFS:u1:x:401:401
    Length: 41
    Data: *:NFS-DOM-1\u1:0:PCNFS:PCNFS:u1:x:401:401
    Padding: Binary Large Object (3 Bytes)
- Map: 0x1
- UNMMapString: *:NFS-DOM-1\u2:0:PCNFS:PCNFS:u2:x:402:401
    Length: 41
    Data: *:NFS-DOM-1\u2:0:PCNFS:PCNFS:u2:x:402:401
    Padding: Binary Large Object (3 Bytes)
- Map: 0x1
- UNMMapString: *:NFS-DOM-1\u3:0:PCNFS:PCNFS:u3:x:403:402
    Length: 41

```

```

        Data: *:NFS-DOM-1\u3:0:PCNFS:PCNFS:u3:x:403:402
        Padding: Binary Large Object (3 Bytes)
- Map: 0x1
  - UNMMapString: -:NFS-DOM-1\spec:0:PCNFS:PCNFS:spec:x:500:500
    Length: 45
    Data: -:NFS-DOM-1\spec:0:PCNFS:PCNFS:spec:x:500:500
    Padding: Binary Large Object (3 Bytes)
- Map: 0x1
  - UNMMapString: -:NFS-DOM-1\u4:0:PCNFS:PCNFS:u4:x:404:402
    Length: 41
    Data: -:NFS-DOM-1\u4:0:PCNFS:PCNFS:u4:x:404:402
    Padding: Binary Large Object (3 Bytes)
- Map: 0x1
  - UNMMapString: -:NFS-DOM-1\u5:0:PCNFS:PCNFS:u5:x:405:401
    Length: 41
    Data: -:NFS-DOM-1\u5:0:PCNFS:PCNFS:u5:x:405:401
    Padding: Binary Large Object (3 Bytes)
- Map: 0x1
  - UNMMapString: -:NFS-DOM-1\u6:0:PCNFS:PCNFS:u6:x:406:402
    Length: 41
    Data: -:NFS-DOM-1\u6:0:PCNFS:PCNFS:u6:x:406:402
    Padding: Binary Large Object (3 Bytes)

```

4.7 GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the Windows group mapping for **POSIX** group "bin".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 53170, Total IP Length = 88
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 68
- Rpc: Call, Program = mapsvc, Procedure =
  GETNTCREDSFROMUNIXGROUPNAME_PROC
  TransactionID: 1473071442 (0x57CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455 (0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETNTCREDSFROMUNIXGROUPNAME_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: GETNTCREDSFROMUNIXGROUPNAME_PROC Call
- UnixGroupAccount:
  SearchOption: UnixAccountName and ID are both valid
  Reserved: Send as 0x00000000
  ID: 1
- UnixAccountName: 0x1
  - UNMName: bin
    Length: 3
    Data: bin
    Padding: Binary Large Object (1 Bytes)

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **group map** for POSIX group "bin" to Windows group "nfs-dom-1\Domain Admins", illustrated as follows.

```

Frame:

```

```

+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 49784, Total IP Length = 88
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 68
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1473071442 (0x57CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
  - MessageAccepted:
    - Verification:
      Flavor: No Identity Authentication
      AuthDataLength: 0 (0x0)
      AcceptState: Call succeeded
- Unm: GETNIXCREDSFROMUNIXGROUPNAME_PROC Reply
- WindowsCreds:
  Status: 0
  Reserved: Send as 0x00000000
- WindowsAccountName: 0x1
  - UNMWindowsName: NFS-DOM-1\Domain Admins
    Length: 23
    Data: NFS-DOM-1\Domain Admins
    Padding: Binary Large Object (1 Bytes)

```

4.8 GETUNIXCREDSFROMNTGROUPNAME_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** group mapping for the Windows group "nfs-dom-1\g1".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 54821, Total IP Length = 84
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 64
- Rpc: Call, Program = mapsvc, Procedure =
  GETUNIXCREDSFROMNTGROUPNAME_PROC
  TransactionID: 1489848658 (0x58CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455 (0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETUNIXCREDSFROMNTGROUPNAME_PROC
  - Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: GETUNIXCREDSFROMNTGROUPNAME_PROC Call
- WindowsGroupAccountName:
  - WindowsAccountName: 0x1
    - UNMName: nfs-dom-1\g1
      Length: 12
      Data: nfs-dom-1\g1

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **group map** for Windows group "nfs-dom-1\g1" to the POSIX group "g1", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 50256, Total IP Length = 68
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 48
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1489848658 (0x58CD4952)

```

```

    MessageType: Reply
-   ServiceReply:
    ReplyStatus: Message accepted
-   MessageAccepted:
    - Verification:
        Flavor: No Identity Authentication
        AuthDataLength: 0 (0x0)
        AcceptState: Call succeeded
-   Unm: GETUNIXCREDSFROMNTGROUPNAME_PROC Reply
-   UnixCreds:
    - UnixAccountName: 0x1
    - UNMName: g1
        Length: 2
        Data: g1
        Padding: Binary Large Object (2 Bytes)
    ID: 401
    GidCount: 0

```

4.9 GETUNIXCREDSFROMNTUSERSID_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** credentials for the Windows user **SID** "S-1-5-21-3994172400-2625080034-4079281819-500" that represents Windows user account "nfs-dom-1\administrator".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 51864, Total IP Length = 100
+ Udp: SrcPort = 1013, DstPort = UNM(819), Length = 80
- Rpc: Call, Program = mapsvc, Procedure =
  GETUNIXCREDSFROMNTUSERSID_PROC
  TransactionID: 1238234037 (0x49CDF3B5)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETUNIXCREDSFROMNTUSERSID_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: GETUNIXCREDSFROMNTUSERSID_PROC Call
- Sid:
  Sidlength: 28
  SID: 01 05 00 00 00 00 00 05 15 00 00 00 F0 3B 12 EE
      E2 8A 77 9C 9B E6 24 F3 F4 01 00 00

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the POSIX credentials for the mapped UNIX account that corresponds to Windows user "nfs-dom-1\Administrator" as POSIX user "root", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 14698, Total IP Length = 76
+ Udp: SrcPort = UNM(819), DstPort = 1013, Length = 56
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1238234037 (0x49CDF3B5)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:

```

```

- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
  AcceptState: Call succeeded
- Unm: GETUNIXCREDSFROMNTUSERSID_PROC Reply
- UnixCreds:
  - UnixAccountName: 0x1
  - UNMName: root
    Length: 4
    Data: root
  ID: 0
  GidCount: 2
- GID:
  GID: 1
  GID: 1

```

4.10 DUMPALLMAPSW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting an enumeration of all user maps (**PrincipalType=0**) starting at index zero (**MapRecordIndex=0**).

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 59252, Total IP Length = 76
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 56
- Rpc: Call, Program = mapsvc, Procedure = DUMPALLMAPSW_PROC
  TransactionID: 1590511954 (0x5ECD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: DUMPALLMAPSW PROC
  - Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: DUMPALLMAPSW_PROC Call
- EnumCursor:
  PrincipalType: Enumerate user account mappings
  MapRecordIndex: 0

```

The User Name Mapping Protocol service on the server responds with a listing of advanced and simple user mappings in the database. The response packet includes a sequence number that indicates the version for the current set of account mappings, a record count that indicates the number of mappings returned as a part of the current packet payload, the total number of maps in the database of the requested types, and finally, the individual maps themselves.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 55477, Total IP Length = 424
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 404
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1590511954 (0x5ECD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)

```

```

    AcceptState: Call succeeded
- Unm: DUMPALLMAPSW_PROC Reply
- MappingW:
  - Token:
    CurrentVersionTokenLowPart: 19924186
    CurrentVersionTokenHighPart: 0
    MappingRecordCount: 8
    TotalMappingRecordCount: 8
  - Map:
    - WindowsAccountName: 0x1
      - UNMNameW: nfs-dom-1\administrator
        Length: 46
        Data: nfs-dom-1\administrator
        Padding: Binary Large Object (2 Bytes)
    - UnixAccountName: 0x1
      - UNMNameW: root
        Length: 8
        Data: root
      ID: 0
    - Map:
      - WindowsAccountName: 0x1
        - UNMNameW: NFS-DOM-1\u1
          Length: 24
          Data: NFS-DOM-1\u1
      - UnixAccountName: 0x1
        - UNMNameW: u1
          Length: 4
          Data: u1
      ID: 401
    - Map:
      - WindowsAccountName: 0x1
        - UNMNameW: NFS-DOM-1\u2
          Length: 24
          Data: NFS-DOM-1\u2
      - UnixAccountName: 0x1
        - UNMNameW: u2
          Length: 4
          Data: u2
      ID: 402
    - Map:
      - WindowsAccountName: 0x1
        - UNMNameW: NFS-DOM-1\u3
          Length: 24
          Data: NFS-DOM-1\u3
      - UnixAccountName: 0x1
        - UNMNameW: u3
          Length: 4
          Data: u3
      ID: 403
    - Map:
      - WindowsAccountName: 0x1
        - UNMNameW: NFS-DOM-1\spec
          Length: 28
          Data: NFS-DOM-1\spec
      - UnixAccountName: 0x1
        - UNMNameW: spec
          Length: 8
          Data: spec
      ID: 500
    - Map:
      - WindowsAccountName: 0x1
        - UNMNameW: NFS-DOM-1\u4
          Length: 24
          Data: NFS-DOM-1\u4
      - UnixAccountName: 0x1
        - UNMNameW: u4
          Length: 4
          Data: u4
      ID: 404

```

```

- Map:
  - WindowsAccountName: 0x1
    - UNMNameW: NFS-DOM-1\u5
      Length: 24
      Data: NFS-DOM-1\u5
  - UnixAccountName: 0x1
    - UNMNameW: u5
      Length: 4
      Data: u5
    ID: 405
- Map:
  - WindowsAccountName: 0x1
    - UNMNameW: NFS-DOM-1\u6
      Length: 24
      Data: NFS-DOM-1\u6
  - UnixAccountName: 0x1
    - UNMNameW: u6
      Length: 4
      Data: u6
    ID: 406

```

4.11 DUMPALLMAPSEXW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting an enumeration of all user maps (**PrincipalType=0**) starting at index zero (**MapRecordIndex=0**).

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 60653, Total IP Length = 76
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 56
- Rpc: Call, Program = mapsvc, Procedure = DUMPALLMAPSEXW_PROC
  TransactionID: 1607289170 (0x5FCD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: DUMPALLMAPSEXW_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: DUMPALLMAPSEXW_PROC Call
  - EnumCursor:
    PrincipalType: Enumerate user account mappings
    MapRecordIndex: 0

```

The User Name Mapping Protocol service on the server responds with a listing of advanced and simple user mappings in the database. The response packet includes a sequence number that indicates the version for the current set of account mappings, a record count that indicates the number of mappings returned as a part of the current packet payload, the total number of maps in the database of the requested types, and finally, the individual maps themselves.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 56145, Total IP Length = 800
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 780
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1607289170 (0x5FCD4952)
  MessageType: Reply
- ServiceReply:

```

```

    ReplyStatus: Message accepted
  - MessageAccepted:
    - Verification:
      Flavor: No Identity Authentication
      AuthDataLength: 0 (0x0)
      AcceptState: Call succeeded
  - Unm: DUMPALLMAPSEXW_PROC Reply
  - MapsW:
    - Token:
      CurrentVersionTokenLowPart: 19924186
      CurrentVersionTokenHighPart: 0
      MappingRecordCount: 8
      TotalMappingRecordCount: 8
    - Map: 0x1
      - UNMMapStringW: *:nfs-dom-1\administrator:0:PCNFS:PCNFS:
        root:x:0:1:1
        Length: 104
        Data: *:nfs-dom-1\administrator:0:PCNFS:PCNFS:root:x:0:1:1
    - Map: 0x1
      - UNMMapStringW: *:NFS-DOM-1\u1:0:PCNFS:PCNFS:u1:x:401:401
        Length: 82
        Data: *:NFS-DOM-1\u1:0:PCNFS:PCNFS:u1:x:401:401
        Padding: Binary Large Object (2 Bytes)
    - Map: 0x1
      - UNMMapStringW: *:NFS-DOM-1\u2:0:PCNFS:PCNFS:u2:x:402:401
        Length: 82
        Data: *:NFS-DOM-1\u2:0:PCNFS:PCNFS:u2:x:402:401
        Padding: Binary Large Object (2 Bytes)
    - Map: 0x1
      - UNMMapStringW: *:NFS-DOM-1\u3:0:PCNFS:PCNFS:u3:x:403:402
        Length: 82
        Data: *:NFS-DOM-1\u3:0:PCNFS:PCNFS:u3:x:403:402
        Padding: Binary Large Object (2 Bytes)
    - Map: 0x1
      - UNMMapStringW: -:NFS-DOM-1\spec:0:PCNFS:PCNFS:spec:x:500:500
        Length: 90
        Data: -:NFS-DOM-1\spec:0:PCNFS:PCNFS:spec:x:500:500
        Padding: Binary Large Object (2 Bytes)
    - Map: 0x1
      - UNMMapStringW: -:NFS-DOM-1\u4:0:PCNFS:PCNFS:u4:x:404:402
        Length: 82
        Data: -:NFS-DOM-1\u4:0:PCNFS:PCNFS:u4:x:404:402
        Padding: Binary Large Object (2 Bytes)
    - Map: 0x1
      - UNMMapStringW: -:NFS-DOM-1\u5:0:PCNFS:PCNFS:u5:x:405:401
        Length: 82
        Data: -:NFS-DOM-1\u5:0:PCNFS:PCNFS:u5:x:405:401
        Padding: Binary Large Object (2 Bytes)
    - Map: 0x1
      - UNMMapStringW: -:NFS-DOM-1\u6:0:PCNFS:PCNFS:u6:x:406:402
        Length: 82
        Data: -:NFS-DOM-1\u6:0:PCNFS:PCNFS:u6:x:406:402
        Padding: Binary Large Object (2 Bytes)

```

4.12 GETWINDOWSUSERFROMUNIXUSERNAMEW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the Windows user mapping for **POSIX** user "root".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 61716, Total IP Length = 92
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 72
- Rpc: Call, Program = mapsvc, Procedure =
  GETNTCREDSFROMUNIXUSERNAMEW_PROC
  TransactionID: 1624066386 (0x60CD4952)

```

```

    MessageType: Call
- ServiceCall:
    RPCVersionNumber: 2 (0x2)
    ProgramNumber: mapsvc, 351455 (0x00055CDF)
    ProgramVersion: 2 (0x2)
    ProcedureNumber: GETNTCREDSFROMUNIXUSERNAMEW PROC
- Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: GETNTCREDSFROMUNIXUSERNAMEW_PROC Call
- UnixUserW:
    SearchOption: UnixAccountName and ID are both valid
    Reserved: Send as 0x00000000
    ID: 0
- UnixAccountName: 0x1
- UNMNameW: root
    Length: 8
    Data: root

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **user map** for POSIX user "root" to Windows user "nfs-dom-1\Administrator", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 60521, Total IP Length = 112
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 92
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
    TransactionID: 1624066386 (0x60CD4952)
    MessageType: Reply
- ServiceReply:
    ReplyStatus: Message accepted
- MessageAccepted:
- Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
    AcceptState: Call succeeded
- Unm: GETNTCREDSFROMUNIXUSERNAMEW_PROC Reply
- WindowsCredsW:
    Status: 0
    Reserved: Send as 0x00000000
- WindowsAccountName: 0x1
- UNMWindowsNameW: nfs-dom-1\administrator
    Length: 46
    Data: nfs-dom-1\administrator
    Padding: Binary Large Object (2 Bytes)

```

4.13 GETUNIXCREDSFROMNTUSERNAMEW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** user mapping for the Windows user "nfs-dom-1\Administrator".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 62611, Total IP Length = 120
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 100
- Rpc: Call, Program = mapsvc, Procedure =
    GETUNIXCREDSFROMNTUSERNAMEW_PROC
    TransactionID: 1640843602 (0x61CD4952)
    MessageType: Call

```

```

- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETUNIXCREDSFROMNTUSERNAMEW_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: GETUNIXCREDSFROMNTUSERNAMEW_PROC Call
- WindowsUserAccountNameW:
  - WindowsAccountName: 0x1
    - UNMNameW: nfs-dom-1\administrator
      Length: 46
      Data: nfs-dom-1\administrator
      Padding: Binary Large Object (2 Bytes)

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **user map** for Windows user "nfs-dom-1\Administrator" to the POSIX user "root", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 63086, Total IP Length = 80
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 60
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1640843602 (0x61CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
    AcceptState: Call succeeded
- Unm: GETUNIXCREDSFROMNTUSERNAMEW_PROC Reply
- UnixCredsW:
  - UnixAccountName: 0x1
    - UNMNameW: root
      Length: 8
      Data: root
  ID: 0
  GidCount: 2
- GID:
  GID: 1
  GID: 1

```

4.14 AUTHUSINGUNIXCREDSW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** account details for the POSIX user "root" with an empty password.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 64478, Total IP Length = 84
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 64
- Rpc: Call, Program = mapsvc, Procedure = AUTHUSINGUNIXCREDSW_PROC
  TransactionID: 1724729682 (0x66CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)

```

```

    ProgramNumber: mapsvc, 351455 (0x00055CDF)
    ProgramVersion: 2 (0x2)
    ProcedureNumber: AUTHUSINGUNIXCREDSW_PROC
  - Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: AUTHUSINGUNIXCREDSW_PROC Call
  - UnixUserAuthW:
    - UnixUserAccountName: 0x1
      - UNMNameW: root
        Length: 8
        Data: root
    - UnixUserAccountPassword: 0x1
      - UNMNameW:
        Length: 0
        Data:

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the mapped POSIX account details for the user "root", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 5741, Total IP Length = 76
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 56
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1724729682 (0x66CD4952)
  MessageType: Reply
  - ServiceReply:
    ReplyStatus: Message accepted
  - MessageAccepted:
    - Verification:
      Flavor: No Identity Authentication
      AuthDataLength: 0 (0x0)
      AcceptState: Call succeeded
- Unm: AUTHUSINGUNIXCREDSW_PROC Reply
  - UnixCredsW:
    - UnixAccountPassword: 0x1
      - UNMNameW: x
        Length: 2
        Data: x
      Padding: Binary Large Object (2 Bytes)
    ID: 0
    GidCount: 2
  - GID:
    GID: 1
    GID: 1

```

4.15 GETWINDOWSGROUPFROMUNIXGROUPNAMEW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the Windows group mapping for **POSIX** group "g1".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 65220, Total IP Length = 88
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 68
- Rpc: Call, Program = mapsvc, Procedure =
  GETNTCREDSFROMUNIXGROUPNAMEW_PROC
  TransactionID: 1741506898 (0x67CD4952)
  MessageType: Call

```

```

- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)
  ProcedureNumber: GETNTCREDSFROMUNIXGROUPNAMEW_PROC
- Credential: No Identity Authentication
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Verification:
  Flavor: No Identity Authentication
  AuthDataLength: 0 (0x0)
- Unm: GETNTCREDSFROMUNIXGROUPNAMEW_PROC Call
- UnixGroupAccountW:
  SearchOption: UnixAccountName and ID are both valid
  Reserved: Send as 0x00000000
  ID: 401
- UnixAccountName: 0x1
  - UNMNameW: g1
    Length: 4
    Data: g1

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **group map** for POSIX group "bin" to Windows group "nfs-dom-1\Domain Admins", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 8813, Total IP Length = 88
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 68
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1741506898 (0x67CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
- MessageAccepted:
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
    AcceptState: Call succeeded
- Unm: GETNTCREDSFROMUNIXGROUPNAMEW_PROC Reply
- WindowsCredsW:
  Status: 0
  Reserved: Send as 0x00000000
- WindowsAccountName: 0x1
  - UNMWindowsNameW: NFS-DOM-1\g1
    Length: 24
    Data: NFS-DOM-1\g1

```

4.16 GETUNIXCREDSFROMNTGROUPNAMEW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** group mapping for the Windows group "nfs-dom-1\Domain Admins".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 2126, Total IP Length = 120
+ Udp: SrcPort = 965, DstPort = UNM(819), Length = 100
- Rpc: Call, Program = mapsvc, Procedure =
  GETUNIXCREDSFROMNTGROUPNAMEW_PROC
  TransactionID: 1758284114 (0x68CD4952)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)

```

```

    ProgramNumber: mapsvc, 351455(0x00055CDF)
    ProgramVersion: 2 (0x2)
    ProcedureNumber: GETUNIXCREDSFROMNTGROUPNAMEW_PROC
  - Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: GETUNIXCREDSFROMNTGROUPNAMEW_PROC Call
  - WindowsGroupNameW:
    - WindowsAccountName: 0x1
      - UNMNameW: nfs-dom-1\Domain Admins
        Length: 46
        Data: nfs-dom-1\Domain Admins
        Padding: Binary Large Object (2 Bytes)

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the **group map** for Windows group "nfs-dom-1\Domain Adminis" to the POSIX group "bin", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 14900, Total IP Length = 72
+ Udp: SrcPort = UNM(819), DstPort = 965, Length = 52
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1758284114 (0x68CD4952)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
  - MessageAccepted:
    - Verification:
      Flavor: No Identity Authentication
      AuthDataLength: 0 (0x0)
      AcceptState: Call succeeded
- Unm: GETUNIXCREDSFROMNTGROUPNAMEW_PROC Reply
  - UnixCredsW:
    - UnixAccountName: 0x1
      - UNMNameW: bin
        Length: 6
        Data: bin
        Padding: Binary Large Object (2 Bytes)
    ID: 1
    GidCount: 0

```

4.17 GETUNIXCREDSFROMNTUSERSIDW_PROC

The client sends a **SUNRPC** packet to the User Name Mapping Protocol service requesting the **POSIX** credentials for the Windows user **SID** "S-1-5-21-3994172400-2625080034-4079281819-500" representing Windows user account "nfs-dom-1\administrator".

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 50594, Total IP Length = 100
+ Udp: SrcPort = 1013, DstPort = UNM(819), Length = 80
- Rpc: Call, Program = mapsvc, Procedure =
  GETUNIXCREDSFROMNTUSERSIDW_PROC
  TransactionID: 1221456821 (0x48CDF3B5)
  MessageType: Call
- ServiceCall:
  RPCVersionNumber: 2 (0x2)
  ProgramNumber: mapsvc, 351455(0x00055CDF)
  ProgramVersion: 2 (0x2)

```

```

    ProcedureNumber: GETUNIXCREDSFROMNTUSERSIDW PROC
  - Credential: No Identity Authentication
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
  - Verification:
    Flavor: No Identity Authentication
    AuthDataLength: 0 (0x0)
- Unm: GETUNIXCREDSFROMNTGROUPNAMEW_PROC Call
  - WindowsUserNameW:
    - WindowsAccountName: 0x1
    - UNMNameW:
      Length: 28
      Data: 01 05 00 00 00 00 00 05 15 00 00 00 F0 3B 12 EE
           E2 8A 77 9C 9B E6 24 F3 F4 01 00 00

```

The User Name Mapping Protocol service on the server responds with a SUNRPC response packet with the POSIX credentials for the mapped **UNIX** account corresponding to Windows user "nfs-dom-1\Administrator" as POSIX user "root", illustrated as follows.

```

Frame:
+ Ethernet: Etype = Internet IP (IPv4)
+ Ipv4: Next Protocol = UDP, Packet ID = 13116, Total IP Length = 80
+ Udp: SrcPort = UNM(819), DstPort = 1013, Length = 60
- Rpc: Reply, Status = Message accepted, Detail = Call succeeded
  TransactionID: 1221456821 (0x48CDF3B5)
  MessageType: Reply
- ServiceReply:
  ReplyStatus: Message accepted
  - MessageAccepted:
    - Verification:
      Flavor: No Identity Authentication
      AuthDataLength: 0 (0x0)
      AcceptState: Call succeeded
- Unm: GETUNIXCREDSFROMNTGROUPNAMEW_PROC Reply
  - UnixCredsW:
    - UnixAccountName: 0x1
    - UNMNameW: root
      Length: 8
      Data: root
    ID: 0
    GidCount: 2
  - GID:
    GID: 1
    GID: 1

```

5 Security

The User Name Mapping Protocol accepts requests with SUNRPC authentication level AUTH_NULL.

5.1 Security Considerations for Implementers

None.

5.2 Index of Security Parameters

None.

6 Appendix A: Full SunRPC IDL

This IDL section excludes the following procedures, which need to be coded separately because the IDL is unable to describe the returned data types. Sample code for the required structure definitions and encode/decode routines can be found in section [Z](#).

Version 1

- DUMPALLMAPS_PROC (procedure 4)
- DUMPALLMAPSEX_PROC (procedure 6)

Version 2

- DUMPALLMAPS_PROC (procedure 4)
- DUMPALLMAPSEX_PROC (procedure 6)
- DUMPALLMAPSW_PROC (procedure 10)
- DUMPALLMAPSEXW_PROC (procedure 11)

```
const MAXNAMELEN = 128;
const MAXNAMELENx2 = 256;
const MAXLINELEN = 256;
const MAXLINELENx2 = 512;
const MAXGIDS = 32;
const MAXSIDLEN = 72;

typedef opaque MapSvrMBCSNameString<MAXNAMELEN>;
typedef opaque MapSvrUnicodeNameString<MAXNAMELENx2>;
typedef opaque MapSvrMBCSWindowsNameString<MAXLINELEN>;
typedef opaque MapSvrUnicodeWindowsNameString<MAXLINELENx2>;
typedef opaque MapSvrMBCSMapString<MAXLINELEN>;
typedef opaque MapSvrUnicodeMapString<MAXLINELENx2>;

struct unix_account {
    long SearchOption;
    long Reserved;
    long ID;
    MapSvrMBCSNameString UnixAccountName;
};

struct unix_accountW {
    long SearchOption;
    long Reserved;
    long ID;
    MapSvrUnicodeNameString UnixAccountName;
};

struct unix_user_auth {
    MapSvrMBCSNameString UnixUserAccountName;
    MapSvrMBCSNameString UnixUserAccountPassword;
};

struct unix_user_authW {
    MapSvrUnicodeNameString UnixUserAccountName;
    MapSvrUnicodeNameString UnixUserAccountPassword;
};

struct windows_creds {
    long Status;
    long Reserved;
    MapSvrMBCSWindowsNameString WindowsAccountName;
};
```

```

struct windows_credsW {
    long Status;
    long Reserved;
    MapSvrUnicodeWindowsNameString WindowsAccountName;
};

struct windows_account {
    MapSvrMBCSNameString WindowsAccountName;
};

struct windows_accountW {
    MapSvrUnicodeNameString WindowsAccountName;
};

struct unix_auth {
    MapSvrMBCSNameString UnixAccountPassword;
    long ID;
    long GIDArray<MAXGIDS>;
};

struct unix_authW {
    MapSvrUnicodeNameString UnixAccountPassword;
    long ID;
    long GIDArray<MAXGIDS>;
};

struct unix_creds {
    MapSvrMBCSNameString UnixAccountName;
    long ID;
    long GIDArray<MAXGIDS>;
};

struct unix_credsW {
    MapSvrUnicodeNameString UnixAccountName;
    long ID;
    long GIDArray<MAXGIDS>;
};

struct dump_map_req {
    long PrincipalType;
    long MapRecordIndex;
};

struct sequence_number {
    long CurrentVersionTokenLowPart;
    long CurrentVersionTokenHighPart;
};

struct mapping_record {
    MapSvrMBCSNameString WindowsAccountName;
    MapSvrMBCSNameString UnixAccountName;
    long ID;
};

struct sid {
    char SID<MAXSIDLEN>;
};

struct mapping_recordW {
    MapSvrUnicodeNameString WindowsAccountName;
    MapSvrUnicodeNameString UnixAccountName;
    long ID;
};

program MAPPROG {
    version MAPVERS_V1 {
        void
        MAPPROC_NULL(void) = 0;
    }
}

```

```

windows_creds
GETWINDOWSCREDSFROMUNIXUSERNAME_PROC(unix_account)= 1;

unix_creds
GETUNIXCREDSFROMNTUSERNAME_PROC(windows_account) = 2;

unix_auth
AUTHUSINGUNIXCREDS_PROC(unix_user_auth) = 3;

sequence number
GETCURRENTVERSIONTOKEN_PROC(sequence_number) = 5;

windows_creds
GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC(unix_account)= 7;

unix_creds
GETUNIXCREDSFROMNTGROUPNAME_PROC(windows_account) = 8;
} = 1;
} = 351455;

program MAPPROC {
    version MAPVERS V2 {

        void
        MAPPROC_NULL(void) = 0;

        windows_creds
        GETWINDOWSCREDSFROMUNIXUSERNAME_PROC(unix_account)= 1;

        unix_creds
        GETUNIXCREDSFROMNTUSERNAME_PROC(windows_account) = 2;

        unix_auth
        AUTHUSINGUNIXCREDS_PROC(unix_user_auth) = 3;

        sequence number
        GETCURRENTVERSIONTOKEN_PROC(sequence_number) = 5;

        windows_creds
        GETWINDOWSGROUPFROMUNIXGROUPNAME_PROC(unix_account)= 7;

        unix_creds
        GETUNIXCREDSFROMNTGROUPNAME_PROC(windows_account) = 8;

        unix_creds
        GETUNIXCREDSFROMNTUSERSID_PROC(sid) = 9;

        windows_credsW
        GETWINDOWSUSERFROMUNIXUSERNAMEW_PROC(unix_accountW)=12;

        unix_credsW
        GETUNIXCREDSFROMNTUSERNAMEW_PROC(windows_accountW)= 13;

        unix_authW
        AUTHUSINGUNIXCREDSW_PROC(unix_user_authW) = 14;

        windows_credsW
        GETWINDOWSGROUPFROMUNIXGROUPNAMEW_PROC(unix_accountW)= 15;

        unix_credsW
        GETUNIXCREDSFROMNTGROUPNAMEW_PROC(windows_accountW) = 16;

        unix_credsW
        GETUNIXCREDSFROMNTUSERSIDW_PROC(sid) = 17;
    } = 2;
} = 351455;

```


7 Appendix B: Sample Code to Encode and Decode Non-XDR-Compliant Data Types

Interpret the sample code in the following sections as being written in the C programming language.

7.1 Header File Content

```
struct mapping {
    sequence_number Token;
    uint MappingRecordCount;
    uint TotalMappingRecordCount;
    mapping_record *MapArray;
};
typedef struct mapping mapping;

struct maps {
    sequence_number Token;
    uint MappingRecordCount;
    uint TotalMappingRecordCount;
    MapSvrMBCSMapString *MapArray;
};
typedef struct maps maps;

struct mappingW {
    sequence_number Token;
    uint MappingRecordCount;
    uint TotalMappingRecordCount;
    mapping_recordW *MapArray;
};
typedef struct mappingW mappingW;

struct mapsW {
    sequence_number Token;
    uint MappingRecordCount;
    uint TotalMappingRecordCount;
    MapSvrUnicodeMapString *MapArray;
};
typedef struct mapsW mapsW;

#define DUMPALLMAPS_PROC 4
extern mapping * dumpallmaps_proc 1(dump_map_req *, CLIENT *);
extern mapping * dumpallmaps_proc 1_svc(dump_map_req *,
                                         struct svc_req *);

#define DUMPALLMAPSEX_PROC 6
extern maps * dumpallmapsex_proc 1(dump_map_req *, CLIENT *);
extern maps * dumpallmapsex_proc 1_svc(dump_map_req *,
                                         struct svc_req *);

#define DUMPALLMAPS_PROC 4
extern mapping * dumpallmaps_proc 2(dump_map_req *, CLIENT *);
extern mapping * dumpallmaps_proc 2_svc(dump_map_req *,
                                         struct svc_req *);

#define DUMPALLMAPSEX_PROC 6
extern maps * dumpallmapsex_proc 2(dump_map_req *, CLIENT *);
extern maps * dumpallmapsex_proc 2_svc(dump_map_req *,
                                         struct svc_req *);

#define DUMPALLMAPSW_PROC 10
extern mappingW * dumpallmapsw_proc 2(dump_map_req *, CLIENT *);
extern mappingW * dumpallmapsw_proc 2_svc(dump_map_req *,
                                         struct svc_req *);
```

```

#define DUMPALLMAPSEXW_PROC 11
extern mapsW * dumpallmapsexw_proc_2(dump_map_req *, CLIENT *);
extern mapsW * dumpallmapsexw_proc_2_svc(dump_map_req *,
                                         struct svc_req *);

extern bool_t xdr_mapping(XDR *, mapping*);
extern bool_t xdr_maps(XDR *, maps*);
extern bool_t xdr_mappingW(XDR *, mappingW*);
extern bool_t xdr_mapsW(XDR *, mapsW*);

```

7.2 Encode/Decode Routines for Non-XDR Data Types Using XDR Primitives

```

bool_t
xdr_mapping(register XDR *xdrs, mapping *objp)
{
    if (!xdr_sequence_number(xdrs, &objp->Token))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->MappingRecordCount))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->TotalMappingRecordCount))
        return (FALSE);

    objp->MapArray = (mapping_record *) malloc (
        objp->MappingRecordCount * sizeof (mapping_record));

    if (!objp->MapArray)
        return (FALSE);

    if (!xdr_vector(xdrs,
                    (char *)objp->MapArray,
                    objp->MappingRecordCount,
                    sizeof (mapping_record),
                    (xdrproc_t) xdr_mapping_record))
        return (TRUE);
}

bool_t
xdr_maps(register XDR *xdrs, maps *objp)
{
    if (!xdr_sequence_number(xdrs, &objp->Token))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->MappingRecordCount))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->TotalMappingRecordCount))
        return (FALSE);

    objp->MapArray = (MapSvrMBCSMapString *) malloc (
        objp->MappingRecordCount * sizeof (MapSvrMBCSMapString));

    if (!objp->MapArray)
        return (FALSE);

    if (!xdr_vector(xdrs,
                    (char *)objp->MapArray,
                    objp->MappingRecordCount,
                    sizeof (MapSvrMBCSMapString),
                    (xdrproc_t) xdr_MapSvrMBCSMapString))
        return (FALSE);

    return (TRUE);
}

bool_t
xdr_mappingW(register XDR *xdrs, mappingW *objp)

```

```

{
    if (!xdr_sequence_number(xdrs, &objp->Token))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->MappingRecordCount))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->TotalMappingRecordCount))
        return (FALSE);

    objp->MapArray = (mapping_recordW *) malloc (
        objp->MappingRecordCount * sizeof (mapping_recordW));

    if (!objp->MapArray)
        return (FALSE);

    if (!xdr_vector(xdrs,
                    (char *)objp->MapArray,
                    objp->MappingRecordCount,
                    sizeof (mapping_recordW),
                    (xdrproc_t) xdr_mapping_recordW))
        return (FALSE);

    return (TRUE);
}

bool_t
xdr_mapsW(register XDR *xdrs, mapsW *objp)
{
    if (!xdr_sequence_number(xdrs, &objp->Token))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->MappingRecordCount))
        return (FALSE);
    if (!xdr_u_int(xdrs, &objp->TotalMappingRecordCount))
        return (FALSE);

    objp->MapArray = (MapSvrUnicodeMapString *) malloc (
        objp->MappingRecordCount * sizeof (MapSvrUnicodeMapString));

    if (!objp->MapArray)
        return (FALSE);

    if (!xdr_vector(xdrs,
                    (char *)objp->MapArray,
                    objp->MappingRecordCount,
                    sizeof (MapSvrUnicodeMapString),
                    (xdrproc_t) xdr_MapSvrUnicodeMapString))
        return (FALSE);

    return (TRUE);
}

```

8 Appendix C: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows Server 2003 R2 operating system
- Windows Vista operating system
- Windows Server 2008 operating system
- Windows 7 operating system
- Windows Server 2008 R2 operating system
- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system
- Windows 10 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows 11 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

<1> [Section 1.5](#): The User Name Mapping Protocol server is only available in Windows Server 2003 R2. In Windows, the User Name Mapping Protocol server is used by the "Client for NFS", "Server for NFS", and "Remote Shell Service" components of the "Services for UNIX" product suite and Windows Server 2003 R2. [\[NFSAUTH\]](#) describes how the Windows User Name Mapping Protocol is configured with appropriate Windows and **UNIX** account mappings.

<2> [Section 2.2.2.6](#): Due to an error in Windows Server 2003 R2 User Name Mapping Protocol server, for maps obtained from a NIS service, the value of the **AuthType** field can unpredictably be set to '0' (AUTH_FILE) rather than the expected value of '1' (AUTH_NIS). The other fields in the map are valid.

Note the User Name Mapping Protocol server is only available in Windows Server 2003 R2.

<3> [Section 2.2.4](#): Although the only authentication mechanism defined by the User Name Mapping Protocol is AUTH_NULL (as specified in [\[RFC1057\]](#) section 9.1), in Windows the User Name Mapping

Protocol server returns a **SUNRPC** status of MSG_DENIED with a reject status of AUTH_ERROR, and an authentication status of AUTH_BADCRED, whenever the client IP address does not match the list of trusted client addresses as configured by the administrator.

<4> [Section 2.2.5.4](#): In Windows, the User Name Mapping Protocol server uses only the UNIX user name as search criteria; the password given as input is ignored.

<5> [Section 2.2.5.15](#): In Windows, the User Name Mapping Protocol server uses only the UNIX user name as search criteria; the password given as input is ignored.

<6> [Section 3.2.5.1](#): In Windows, the User Name Mapping Protocol server returns a SUNRPC status of MSG_DENIED with a reject status of AUTH_ERROR and an authentication status of AUTH_BADCRED, whenever the client IP address does not match the list of trusted client addresses, as configured by the administrator. This Windows-specific behavior is documented in [NFSAUTH].

<7> [Section 3.2.5.2.1](#): In Windows, the User Name Mapping Protocol server returns a maximum of 200 mapping records in a SUNRPC packet response.

<8> [Section 3.2.5.2.1](#): In Windows, the User Name Mapping Protocol initializes **CurrentVersionTokenHighPart** and **CurrentVersionTokenLowPart** to a locally unique identifier as returned by the Win32 API **AllocateLocallyUniqueId** call on startup.

<9> [Section 3.2.5.2.2](#): In Windows, the User Name Mapping Protocol server limits the number of mapping records returned by both **DUMPALLMAPS_PROC** and **DUMPALLMAPSW_PROC** in a single SUNRPC packet response to a value set by the registry DWORD value HKLM\System\CurrentControlSet\Services\MapSvc\CurrentVersion\WriteBlock. The default value for this limit is 200. Any changes to this value require a restart of the Windows User Name Mapping Protocol server to become effective.

<10> [Section 3.2.5.2.2](#): In Windows, the User Name Mapping Protocol server limits the size of a single SUNRPC response that uses the UDP transport to 8,800 bytes. There is no such limit when using the TCP transport. For the **DUMPALLMAPSXXX_PROC** procedures, if the number of records to be returned in a single SUNRPC UDP response message cannot be contained in a message of this size, then Windows User Name Mapping Protocol server replies with a SUNRPC message status of MSG_ACCEPTED, with the ACCEPT status set to SYSTEM_ERR.

<11> [Section 3.2.5.2.2](#): In Windows, the User Name Mapping Protocol server ignores the value of *MapRecordIndex* in the **DUMPALLMAPSEX_PROC** and **DUMPALLMAPSEXW_PROC** request, and the response always contains the maps enumerated from index 0. In the response, the **TotalMappingRecordCount** value is set to the sum of the total number of map records held by the server and the value of *MapRecordIndex* in the request.

<12> [Section 3.2.5.2.3](#): In Windows, the User Name Mapping Protocol server changes the 64-bit integer sequence number to a random value every time the account mapping database is updated. This value is unique only within the lifetime of the current server process—its uniqueness can only be guaranteed within the span of a single server process. The server returns this sequence number to the client as **CurrentVersionTokenHighPart** and **CurrentVersionTokenLowPart** in the [GETCURRENTVERSIONTOKEN_PROC](#) response and the **DUMPALLMAPSXXX_PROC** response.

9 Change Tracking

No table of changes is available. The document is either new or has had no changes since its last release.

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